Joint Code-Dimension and k-Galois-Hull Enumerators for Repeated-Root Constacyclic Codes over Square-Free Affine Algebras

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Abstract

Let $\mathcal{A} \cong \prod_{s=1}^{N} K_s$ be a finite commutative square-free affine algebra over $\mathbb{F}_q$ ($q = p^e$, $K_s = \mathbb{F}_{q^{m_s}}$ with $m_s \geq 1$), let $0 \leq k < e$, let $n = n'p^t \geq 1$ with $\gcd(n', p) = 1$ and multiplicity $M = p^t$, and let $\lambda = (\lambda_s)_{s=1}^{N} \in \mathcal{A}^*$ be any unit with unique M-th roots $\lambda_{s,0} \in K_s^*$. We determine the exact bivariate generating polynomial $\mathscr{E}^{(M)}(U, Z) = \sum_C U^{\dim_{\mathbb{F}_q} C} Z^{\dim_{\mathbb{F}_q} \mathrm{Hull}_{\mathcal{A},k}(C)}$ over all $(M+1)^{N_F}$ labeled $\mathcal{A}$-linear λ-constacyclic codes of length n without assuming $m_s = 1$ or $\gcd(\mathrm{ord}(\lambda_s), n') = 1$. Components failing $\lambda_{s,0}^{1+p^{em_s-k}} = 1$ have trivial k-Galois hulls for every code, whereas on compatible components $\dim_{K_s} \mathrm{Hull}_k(C_s(e))$ splits across inverse-Frobenius reciprocal orbits of $x^{n'} - \lambda_{s,0}$ as cyclic sums of $\min\{e_i, M - e_{i+1}\}$. This yields a universal $(M+1) \times (M+1)$ transfer-matrix trace product for $\mathscr{E}^{(M)}(U, Z)$ evaluating exponentially large code spaces in polynomial time and recovering the simple-root theory at $M = 1$. From it we prove: (i) p-power invariance $2^{\sum_s |\mathcal{O}_s|}$ of the k-Galois LCD code count across all $t \geq 0$ for compatible twists; (ii) an unconditional characterisation of the single-orbit support trapezoid $\min\{R, M, aM - R\} \leq H \leq \min\{R, aM - R\}$ and the global Minkowski hull spectrum; (iii) closed rational formulas for $\mathbb{E}[\eta_k]$, $\mathrm{Var}(\eta_k)$ across the correlation regimes $a = 1$, $a = 2$, and $a \geq 3$, $\mathrm{Cov}(\kappa, \eta_k) = 0$, and conditional moments given κ; and (iv) exact enumerators for self-orthogonal, dual-containing, and self-dual codes revealing a sharp $p = 2$ versus odd-p dichotomy. All results are verified independently by finite-field Gram-rank and raw quotient-polynomial computations across characteristics 2, 3, and 5.

Keywords: repeated-root constacyclic codes, k-Galois hulls, square-free affine algebras, exact enumeration, transfer matrices, LCD codes, self-dual codes
2020 MSC: 94B05, 94B15, 05A15, 11T71

1. Introduction

The *hull* $\mathrm{Hull}(C) = C \cap C^{\perp}$ of a linear code C governs two fundamental threads of modern coding theory: at one extreme, linear codes with complementary duals ($\mathrm{Hull}(C) = \{0\}$, LCD

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codes) introduced by Massey [1, 2] satisfy the Gilbert–Varshamov bound [3] and protect against side-channel and fault-injection attacks [4]; at the opposite and intermediate regimes, the hull dimension directly determines the algorithmic complexity of permutation-equivalence and automorphism-group algorithms [5, 6] as well as the number of pre-shared maximally entangled ebits required to build entanglement-assisted quantum error-correcting codes (EAQECCs) [7–10]. Over a finite field of characteristic p, replacing the Euclidean inner product by Fan and Zhang's k-Galois pairing [11] interpolates unifiedly between Euclidean ($k = 0$), Hermitian ($k = d/2$ for even degree d), and intermediate Frobenius dualities [12, 13].

For simple-root constacyclic codes ($\gcd(n, p) = 1$), hull dimensions, average hull dimensions, and enumerative counts over finite fields and ambient rings have been investigated extensively [14–26]. Recently, Debnath, Islam, Martínez-Moro, and Prakash [27] studied k-Galois hulls of λ-constacyclic codes over square-free commutative affine algebras $\mathcal{A}$ in the split case ($m_s = 1$ on every simple component), deriving a repeated-root hull-dimension formula under the cyclotomic hypothesis $\gcd(\mathrm{ord}(\lambda), n') = 1$ and $\mathrm{ord}(\lambda) \mid (1 + p^{e-k})$ [27, Theorem 8], determining the cardinality of the hull-dimension set in two special cases [27, Corollary 3.5], and explicitly posing in their Conclusion [27, p. 23] the open problem of *counting the number of constacyclic codes over $\mathcal{A}$ having each given k-Galois hull dimension.* Related recent works examine first moments (average Galois hull dimensions) over $\mathbb{F}_q$ and split rings $R_{m,q} = \mathbb{F}_q[u]/\langle u^m - u\rangle$ [28] and existence criteria for hull dimensions 1 and 2 over $\mathbb{F}_q$ [29], while classical repeated-root classifications treat restricted coprime parts $n' \in \{1, 2, 2\ell^m\}$ or Euclidean LCD counts [30–36].

In the simple-root regime $\gcd(n, p) = 1$ ($M = 1$), each component quotient $K_s[x]/\langle x^n - \lambda_s\rangle$ is semisimple, every constacyclic code corresponds to a binary factor selection in $\{0, 1\}^{\mathcal{F}_s}$, and the k-Galois hull dimension reduces to a weighted count of circular $0 \to 1$ boundaries in $\{0, 1\}^{a_O}$ governed by a 2×2 transfer matrix (recovered here as the $M = 1$ specialisation in Corollary 4.7(c)). When $p \mid n$, writing $n = n'p^t$ with $\gcd(n', p) = 1$ and $M = p^t \geq p$ changes the algebraic and combinatorial picture qualitatively:

1. Ambient component rings $K_s[x]/\langle x^n - \lambda_s\rangle \cong \prod_{f \in \mathcal{F}_s} K_s[x]/\langle f(x)^M\rangle$ are no longer semisimple products of fields, but products of non-semisimple principal chain rings in which generator exponents vary across the $(M + 1)$-element alphabet $\{0, 1, \ldots, M\}$.
2. The base polynomial whose distinct irreducible factors $\mathcal{F}_s$ index ideals is $x^{n'} - \lambda_{s,0}$ rather than $x^{n'} - \lambda_s$, where $\lambda_{s,0} \in K_s^*$ is the unique p^t-th root of λ_s.
3. Simple-root circular run combinatorics on $\{0, 1\}^{a_O}$ breaks down and is replaced by cyclic sums of pairwise coupled edge minima $\min\{e_i, M - e_{i+1}\}$, creating flat nonzero hull floors at height $w_O M$, three distinct variance regimes ($a_O = 1$, $a_O = 2$, $a_O \geq 3$), and new self-dual codes on odd-length orbits in characteristic $p = 2$.

Main contributions

Figure 1 summarises our end-to-end algebraic and enumerative pipeline, and Table 1 contrasts our scope and results against the closest existing literature. Specifically, for arbitrary $n = n'p^t$ ($t \geq 0$, $M = p^t$), arbitrary non-split component degrees $m_s \geq 1$, arbitrary Galois twists $0 \leq k < e$, and arbitrary units $\lambda \in \mathcal{A}^*$ (with zero coprimality assumption between $\mathrm{ord}(\lambda_s)$ and n'):

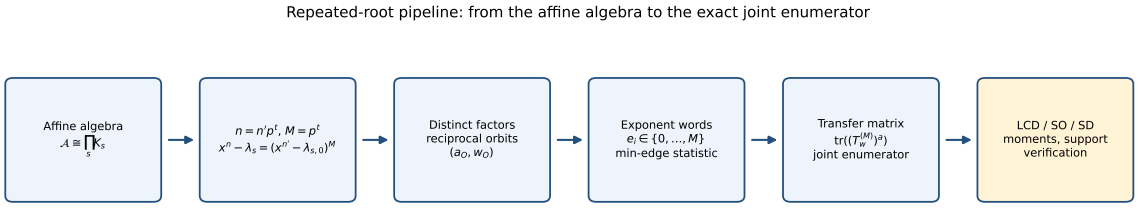


Figure 1: Overview of the exact repeated-root framework: CRT decomposition of the square-free affine algebra $\mathcal{A} \cong \prod_{s=1}^{N} K_s$, pure-power reduction $x^n - \lambda_s = (x^{n'} - \lambda_{s,0})^M$, partition into $\tau_{s,k}$-orbits (a_O, w_O), cyclic edge-minimum hull statistic on $\{0, \ldots, M\}^{a_O}$, $(M+1) \times (M+1)$ transfer-matrix trace product, and exact structural/statistical consequences.

1. **Repeated-root k-Galois duality and general twists (Section 3).** We prove directly, without any square-freeness or order-coprimality hypothesis, that $D_k(C_s(e)) = \langle h_e^{\#_{s,k}}(x) \rangle$ (Theorem 3.4), prove that twist compatibility is invariant under p^t-th roots ($\lambda_{s,0}^{1+p^{d_s-k}} = 1 \iff \lambda_s^{1+p^{d_s-k}} = 1$, Proposition 3.5), completely resolve incompatible components and mixed units $\lambda \in \mathcal{A}^*$ (Proposition 3.8), and establish full convention invariance of all dimension enumerators (Remark 3.10).
2. **Single-code cyclic edge minimum and exact joint transfer-matrix enumerator (Section 4).** On every compatible component, we express $\mathrm{Hull}_k(C_s(e))$ and $\dim_{K_s} \mathrm{Hull}_k(C_s(e))$ via cyclic sums of $\min\{e_{O,i}, M - e_{O,i+1}\}$ (Theorem 4.2, reconciling and strictly extending [27, Theorem 8]), and prove the universal trace formula $\mathscr{E}^{(M)}(U, Z) = \prod_{s,O} \mathrm{tr}(T_{w_O}^{(M)}(U, Z)^{a_O}) \cdot \prod_{s \in S_{\mathrm{inc}}, f} \frac{1 - U^{(M+1) m_s \deg f}}{1 - U^{m_s \deg f}}$ (Theorem 4.6), answering the open enumeration problem of [27, p. 23] jointly with code dimension $\kappa(C)$ and reducing at $M = 1$ to an explicit circular-run binomial formula for the simple-root regime (Corollary 4.7(c)).
3. **LCD p-power invariance, exact support trapezoid, closed moments, and extremal classes (Section 5).** We prove that $T_1^{(M)}(1, 0)$ is a rank-2 matrix with unipotent Jordan block $\left(\begin{smallmatrix} 1 & M \\ 0 & 1 \end{smallmatrix}\right)$, locking the compatible k-Galois LCD count at $2^{\sum_s |\mathcal{O}_s|}$ independently of $t = v_p(n)$ (Theorem 5.2); prove unconditionally that each single-orbit support is the exact lattice trapezoid $\min\{R, M, aM - R\} \le H \le \min\{R, aM - R\}$ of cardinality $aM + 1 + \lfloor (a-2)^2 M^2/4 \rfloor$ ($a \ge 2$), yielding the global Minkowski hull spectrum (Theorem 5.4 and Corollary 5.5); derive exact rational formulas for $\mathbb{E}[\eta_k]$ (recovering the repeated-root first moments of [28] as a one-line special case; e.g. for $q = 2^6$, $n = 28 = 7 \cdot 4$, $k = 3$, (5.9) immediately gives $\frac{4}{5} + 6 \cdot \frac{6}{5} = 8$), $\mathrm{Var}(\eta_k)$ across all three regimes $a = 1$ (even/odd M), $a = 2$, and $a \ge 3$, $\mathrm{Cov}(\kappa, \eta_k) = 0$, and conditional moments $\mathbb{E}[\eta_k \mid \kappa]$, $\mathrm{Var}(\eta_k \mid \kappa)$ (Theorem 5.6 and Proposition 5.7); and give exact enumerators for self-orthogonal, dual-containing, and self-dual codes (Theorem 5.8).
4. **Polynomial-time algorithms and first-principles verification (Section 6).** Proposition 6.1 evaluates spaces of size $5^{31} \approx 4.66 \times 10^{21}$ in $< 0.3\,\mathrm{s}$, while every formula is verified independently by finite-field Gram-rank and raw quotient-polynomial linear algebra across characteristics 2, 3, and 5.

Table 1: Comparison of scope and results with closest existing works on hulls of constacyclic codes.

Reference	Ambient ring	Length n	Pairing	Twist λ	Hull distribution	Moments, support & extremal counts
Sangwisut et al. [15]	$\mathbb{F}_q$	$p \nmid n$	Euclidean, Hermitian	$\lambda \in \{\pm 1\}$	Marginal counts per η	Mean only
Pang–Zhu–Li [36]; Chen et al. [35]	$\mathbb{F}_q$	$n'p^t$; $2\ell^m p^t$	Euclidean ($k=0$)	$\lambda = 1$ or restricted	LCD ($\eta = 0$) only	LCD / SD counts
Debnath et al. [19, 20]	$\mathbb{F}_q$	$p \nmid n$	k-Galois	Compatible $\lambda \in \mathbb{F}_q^*$ (q restricted)	Marginal counts per η	None
Debnath–Prakash [28]	$\mathbb{F}_q$ and split $R_{m,q}$	General n	k-Galois	$\lambda \in \mathbb{F}_q^*$ / split	None	Unconditional mean $\mathbb{E}[\eta_k]$ only
Debnath et al. [27]	Split $\mathcal{A}$ ($m_s = 1$)	$n'p^t$	k-Galois	$\gcd(\mathrm{ord}(\lambda), n') = 1$, compatible	Open problem [27, p. 23]	$\lvert\mathrm{Spec}\rvert$ in 2 cases; LCD criterion
Present work	**General sq.-free $\mathcal{A}$ ($m_s \geq 1$)**	**All $n = n'p^t$ ($M = p^t \geq 1$)**	**k-Galois ($0 \leq k < e$)**	**Arbitrary $\lambda \in \mathcal{A}^*$ (comp. & inc.)**	**Exact $(M+1)\times(M+1)$ trace product** $\mathscr{E}^{(M)}(U, Z)$	**LCD invariance, support trapezoid,** $\mathbb{E}$, Var**, cond. law, SO/DC/SD**

Organisation of the paper

Section 2 establishes the CRT structure, pure-power factorisation, and global exponent-selection bijection. Section 3 proves the repeated-root k-Galois dual generator, twist compatibility, incompatible-twist collapse, and convention invariance. Section 4 proves the single-code cyclic min-formula and the main transfer-matrix Theorem 4.6. Section 5 derives marginals, p-power LCD invariance, the support trapezoid Theorem 5.4, exact moments, and extremal code counts. Section 6 presents polynomial-time algorithms, independent verification, and worked families, and Section 7 records open directions.

2. The repeated-root setting

This section fixes the algebraic framework used throughout the paper: the square-free affine coefficient algebra and its fixed labeled finite-field decomposition (§2.1), the pure-power factorisation of repeated-root constacyclic moduli over each component field (§2.2), the component and global exponent-selection bijections generalising simple-root binary factor selections (§2.3), and our standing assumptions and notation (§2.4).

2.1. Square-free affine algebras and CRT decomposition

Throughout the paper, p is a prime and

$$q = p^e, \qquad e \geq 1,$$

is the order of the base field $\mathbb{F}_q$. Consider a multivariate commutative affine $\mathbb{F}_q$-algebra presented as

$$\mathcal{A} = \mathbb{F}_q[X_1, \ldots, X_\ell] \Big/ \langle t_1(X_1), \ldots, t_\ell(X_\ell) \rangle, \qquad \ell \geq 1, \tag{2.1}$$

where each $t_i(X_i) \in \mathbb{F}_q[X_i]$ is monic, square-free over $\mathbb{F}_q$, and of positive degree. This hypothesis makes $\mathcal{A}$ a reduced finite étale $\mathbb{F}_q$-algebra and yields a canonical product decomposition once the component labels are fixed.

Proposition 2.1 (Square-free component decomposition)**.** *The affine algebra $\mathcal{A}$ in* (2.1) *admits a fixed labeled $\mathbb{F}_q$-algebra isomorphism*

$$\mathcal{A} \cong \prod_{s=1}^{N} K_s, \qquad K_s = \mathbb{F}_{q^{m_s}} = \mathbb{F}_{p^{d_s}}, \qquad d_s = em_s, \quad m_s = [K_s : \mathbb{F}_q] \geq 1, \tag{2.2}$$

together with pairwise orthogonal primitive idempotents $e_1, \ldots, e_N \in \mathcal{A}$ satisfying $e_s e_{s'} = \delta_{s,s'} e_s$, $\sum_{s=1}^{N} e_s = 1_{\mathcal{A}}$, and $e_s \mathcal{A} \cong K_s$ for every $1 \leq s \leq N$.

Proof. Because each t_i is monic of positive degree and square-free over the perfect field $\mathbb{F}_q$, each univariate quotient $\mathbb{F}_q[X_i]/\langle t_i(X_i)\rangle$ decomposes by the Chinese remainder theorem into a finite non-empty product of finite extensions of $\mathbb{F}_q$ and is therefore finite étale over $\mathbb{F}_q$. The presentation (2.1) is the $\mathbb{F}_q$-tensor product of these factors, so $\mathcal{A}$ is itself a non-zero finite étale $\mathbb{F}_q$-algebra and hence isomorphic to a finite product of finite extensions $K_1, \ldots, K_N$ of $\mathbb{F}_q$. Ordering the factors fixes the isomorphism (2.2) and identifies e_s with the tuple having 1_{K_s} in coordinate s and 0 elsewhere. □

We fix the labeled decomposition (2.2) once and for all and write elements of $\mathcal{A}$ in tuple coordinates as $a = (a_s)_{s=1}^{N}$ with $a_s \in K_s$. Notice carefully the scope of the adjective *square-free*: throughout this paper, square-freeness is imposed solely on the defining polynomials $t_1, \ldots, t_\ell$ of the coefficient algebra $\mathcal{A}$ (so that $\mathcal{A}$ has no nilpotent elements), whereas the constacyclic polynomial $x^n - \lambda_s$ in the code length n is allowed to have repeated roots whenever $p \mid n$.

2.2. Repeated-root moduli and pure-power factorisation

Let $n \geq 1$ be an arbitrary integer and write

$$n = n' p^t, \qquad \gcd(n', p) = 1, \qquad t = v_p(n) \geq 0, \qquad M := p^t. \tag{2.3}$$

The simple-root regime $\gcd(n, p) = 1$ corresponds to $t = 0$ ($M = 1$, $n = n'$) and is included uniformly in every statement below.

Fix a global shift unit $\lambda = (\lambda_s)_{s=1}^{N} \in \mathcal{A}^* = \prod_{s=1}^{N} K_s^*$. For each component $1 \leq s \leq N$, define the component constacyclic modulus

$$P_s(x) := x^n - \lambda_s \in K_s[x]. \tag{2.4}$$

Because the absolute Frobenius map $\mathrm{Frob}_s : a \mapsto a^p$ is a field automorphism of $K_s = \mathbb{F}_{p^{d_s}}$, its t-fold iterate $a \mapsto a^{p^t} = a^M$ is also an automorphism of K_s (explicitly inverted by $a \mapsto a^{p^{(-t) \bmod d_s}}$). Every component twist $\lambda_s \in K_s^*$ therefore admits a unique M-th root

$$\lambda_{s,0} := \mathrm{Frob}_s^{-t}(\lambda_s) \in K_s^* \qquad \text{such that} \qquad \lambda_{s,0}^M = \lambda_s. \tag{2.5}$$

Lemma 2.2 (Repeated-root factorisation)**.** *Let $n = n' p^t$ with $\gcd(n', p) = 1$ and $M = p^t$, let $\lambda_s \in K_s^*$, and let $\lambda_{s,0} \in K_s^*$ be defined by* (2.5)*. Denote by $\mathcal{F}_s$ the set of distinct monic irreducible factors of the base polynomial $P_{s,0}(x) := x^{n'} - \lambda_{s,0}$ in $K_s[x]$. Then $P_{s,0}(x)$ is square-free over K_s, every $f \in \mathcal{F}_s$ satisfies $f(0) \neq 0$, $\sum_{f \in \mathcal{F}_s} \deg f = n'$, and*

$$P_s(x) = x^n - \lambda_s = \left(x^{n'} - \lambda_{s,0}\right)^M = \prod_{f \in \mathcal{F}_s} f(x)^M. \tag{2.6}$$

In particular, $\mathcal{F}_s$ is also the set of distinct monic irreducible factors of $P_s(x)$ in $K_s[x]$, and each $f \in \mathcal{F}_s$ occurs in $P_s(x)$ with multiplicity exactly M.

Proof. Existence and uniqueness of $\lambda_{s,0} \in K_s^*$ follow from bijectivity of $a \mapsto a^{p^t}$ on the finite field K_s. Since $K_s[x]$ has characteristic p, the map $g(x) \mapsto g(x)^p$ is a ring endomorphism of $K_s[x]$ sending x to x^p and each constant $c \in K_s$ to c^p; iterating it t times gives

$$\left(x^{n'} - \lambda_{s,0}\right)^M = \left(x^{n'} - \lambda_{s,0}\right)^{p^t} = x^{n'p^t} - \lambda_{s,0}^{p^t} = x^n - \lambda_s = P_s(x),$$

as in the classical repeated-root reduction [30, 31].

Next, the formal derivative of $P_{s,0}(x) = x^{n'} - \lambda_{s,0}$ in $K_s[x]$ is $P'_{s,0}(x) = n'x^{n'-1}$, with $n' \neq 0$ in K_s because $\gcd(n', p) = 1$. In an algebraic closure $\overline{K}_s$, the only possible root of $x^{n'-1}$ is 0, whereas $P_{s,0}(0) = -\lambda_{s,0} \neq 0$. Thus $\gcd(P_{s,0}, P'_{s,0}) = 1$, so $P_{s,0}$ has no repeated roots in $\overline{K}_s$ and factors in the unique factorisation domain $K_s[x]$ into pairwise distinct monic irreducibles $P_{s,0}(x) = \prod_{f \in \mathcal{F}_s} f(x)$. Comparing degrees and constant terms gives $\sum_{f \in \mathcal{F}_s} \deg f = \deg P_{s,0} = n'$ and $\prod_{f \in \mathcal{F}_s} f(0) = -\lambda_{s,0} \neq 0$, whence $f(0) \neq 0$ for every $f \in \mathcal{F}_s$. Raising $P_{s,0}(x) = \prod_{f \in \mathcal{F}_s} f(x)$ to the M-th power gives (2.6); unique factorisation in $K_s[x]$ identifies $\mathcal{F}_s$ with the set of distinct monic irreducible factors of $P_s(x)$ and shows that each occurs to exact power M. □

Remark 2.3. Lemma 2.2 is the entire factorisation input required in this paper. Unlike explicit cyclotomic factorisation formulas over $\mathbb{F}_q$ such as [27, Section 2.4 and Theorem 8], which impose $m_s = 1$ and $\gcd(\mathrm{ord}(\lambda), n') = 1$, or explicit generator classifications for restricted coprime parts $n' \in \{1, 2, 2\ell^m\}$ [33–35], (2.6) holds for arbitrary $\lambda_s \in K_s^*$, arbitrary extension degrees $m_s \geq 1$, and arbitrary coprime part n'. One bookkeeping point is essential throughout: a code of length n is an ideal modulo $P_s(x) = x^n - \lambda_s$, whereas the distinct irreducible factors $\mathcal{F}_s$ are determined by the base polynomial $x^{n'} - \lambda_{s,0}$, and in general $\lambda_{s,0} \neq \lambda_s$ whenever $\lambda_s^{M-1} \neq 1$. Every concrete example below therefore records both $\lambda_{s,0}$ and $\lambda_s = \lambda_{s,0}^M$.

2.3. Constacyclic codes and the exponent-selection bijection

Under the tuple identification $\mathcal{A}^n \cong \prod_{s=1}^N K_s^n$ induced by (2.2), the ambient polynomial module for λ-constacyclic codes is

$$\mathcal{M}_\lambda := \mathcal{A}[x]/\langle x^n - \lambda \rangle \cong \prod_{s=1}^N K_s[x]/\langle P_s(x) \rangle \cong \prod_{s=1}^N K_s^n, \tag{2.7}$$

where the second isomorphism sends a residue class $\sum_{i=0}^{n-1} c_i x^i$ to its coefficient vector $(c_0, c_1, \ldots, c_{n-1})$.

Definition 2.4. For $\lambda \in \mathcal{A}^*$ and $n \geq 1$, the λ-*constacyclic shift* on $\mathcal{A}^n$ is the $\mathcal{A}$-linear map

$$T_\lambda(c_0, c_1, \ldots, c_{n-1}) := (\lambda c_{n-1}, c_0, \ldots, c_{n-2}).$$

A λ-*constacyclic code of length* n *over* $\mathcal{A}$ is an $\mathcal{A}$-submodule $C \subseteq \mathcal{A}^n$ satisfying $T_\lambda(C) \subseteq C$. Under the identification $\mathcal{A}^n \cong \mathcal{M}_\lambda$ in (2.7), T_λ acts as multiplication by x, so the λ-constacyclic codes of length n over $\mathcal{A}$ are precisely the ideals of $\mathcal{M}_\lambda$.

Remark 2.5. Neither Definition 2.4 nor (2.7) requires $\gcd(n, p) = 1$. Throughout this paper, codes are required to be $\mathcal{A}$-submodules of $\mathcal{A}^n$; $\mathbb{F}_q$-linear subspaces closed under T_λ that are not closed under scalar multiplication by $\mathcal{A}$ are outside the scope of our enumeration.

On each component $1 \le s \le N$, an *exponent selection* is a function

$$e^{(s)} = (e_f)_{f \in \mathcal{F}_s} \in \{0, 1, \dots, M\}^{\mathcal{F}_s};$$

when the component index s is clear from context we write $e = (e_f)_{f \in \mathcal{F}_s}$. Associate to $e \in \{0, \dots, M\}^{\mathcal{F}_s}$ the monic generator polynomial, the monic parity-check polynomial, and the component code

$$g_e(x) := \prod_{f \in \mathcal{F}_s} f(x)^{e_f}, \qquad h_e(x) := \frac{P_s(x)}{g_e(x)} = \prod_{f \in \mathcal{F}_s} f(x)^{M - e_f}, \qquad C_s(e) := \langle g_e(x) \rangle \subseteq K_s[x]/\langle P_s(x) \rangle. \tag{2.8}$$

Lemma 2.6 (Component exponent-selection bijection)**.** *For each $1 \le s \le N$, the assignment $e \mapsto C_s(e) = \langle g_e(x) \rangle$ is a bijection from $\{0, \dots, M\}^{\mathcal{F}_s}$ onto the set of λ_s-constacyclic codes of length n over K_s. Moreover,*

$$\dim_{K_s} C_s(e) = n - \sum_{f \in \mathcal{F}_s} e_f \deg f = \sum_{f \in \mathcal{F}_s} (M - e_f) \deg f. \tag{2.9}$$

Proof. Because $K_s[x]$ is a principal ideal domain, the lattice isomorphism theorem puts the ideals of the quotient ring $K_s[x]/\langle P_s(x) \rangle$ in bijection with the ideals of $K_s[x]$ containing $\langle P_s(x) \rangle$, and each such ideal has a unique monic generator dividing $P_s(x)$. Indeed, if $g_1 \mid P_s$ and $g_2 \mid P_s$ are monic and generate the same ideal modulo $\langle P_s \rangle$, then in $K_s[x]$ we have

$$\langle g_1 \rangle = \langle g_1, P_s \rangle = \langle g_2, P_s \rangle = \langle g_2 \rangle,$$

whence $g_1 = g_2$ by monicity. By Lemma 2.2, unique factorisation in $K_s[x]$ puts the monic divisors of $P_s(x) = \prod_{f \in \mathcal{F}_s} f(x)^M$ in bijection with tuples $e = (e_f)_{f \in \mathcal{F}_s} \in \{0, \dots, M\}^{\mathcal{F}_s}$ via $e \mapsto g_e$.

Finally, $K_s[x]/\langle P_s \rangle \big/ \langle g_e \rangle \cong K_s[x]/\langle g_e \rangle$ has K_s-dimension $\deg g_e = \sum_{f \in \mathcal{F}_s} e_f \deg f$, so $\dim_{K_s} C_s(e) = n - \deg g_e$. Since $n = Mn' = M \sum_{f \in \mathcal{F}_s} \deg f$ by Lemma 2.2, subtracting $\sum_{f \in \mathcal{F}_s} e_f \deg f$ yields the second expression in (2.9). □

Define the global exponent-selection space, total distinct-factor count, and ambient $\mathbb{F}_q$-dimension by

$$\mathscr{C}^{(M)} := \prod_{s=1}^{N} \{0, 1, \dots, M\}^{\mathcal{F}_s}, \qquad N_F := \sum_{s=1}^{N} |\mathcal{F}_s|, \qquad W := n \sum_{s=1}^{N} m_s. \tag{2.10}$$

For $e = (e^{(s)})_{s=1}^{N} \in \mathscr{C}^{(M)}$, define the global product code

$$C(e) := \prod_{s=1}^{N} C_s\Big(e^{(s)}\Big) \subseteq \prod_{s=1}^{N} K_s^n \cong \mathcal{A}^n, \tag{2.11}$$

generated in $\mathcal{M}_\lambda$ by the tuple element $g_e := (g_{e^{(s)}})_{s=1}^{N}$.

Lemma 2.7 (Global exponent-selection bijection)**.** *Every λ-constacyclic code $C \subseteq \mathcal{A}^n$ decomposes uniquely as $C = \bigoplus_{s=1}^{N} e_s C$, where each $e_s C$ is an ideal of $e_s \mathcal{M}_\lambda \cong K_s[x]/\langle P_s(x)\rangle$. Consequently, the map*

$$\mathscr{C}^{(M)} \longrightarrow \{\lambda\text{-constacyclic codes of length } n \text{ over } \mathcal{A}\}, \qquad e \longmapsto C(e),$$

is a bijection. In particular, the total number of labeled λ-constacyclic codes of length n over $\mathcal{A}$ is

$$\left|\mathscr{C}^{(M)}\right| = (M+1)^{N_F}, \tag{2.12}$$

and the global $\mathbb{F}_q$-dimension of $C(e)$ is

$$\kappa\Big(C(e)\Big) := \dim_{\mathbb{F}_q} C(e) = \sum_{s=1}^{N} m_s \dim_{K_s} C_s\Big(e^{(s)}\Big) = \sum_{s=1}^{N} \sum_{f \in \mathcal{F}_s} m_s (\deg f)\,(M - e_f). \tag{2.13}$$

Proof. Let $e_1, \ldots, e_N \in \mathcal{A}$ be the orthogonal primitive idempotents of Proposition 2.1. For any ideal $C \subseteq \mathcal{M}_\lambda$, $\mathcal{A}$-linearity gives $e_s C \subseteq C$, and $\sum_{s=1}^{N} e_s = 1_{\mathcal{A}}$ with $e_s e_{s'} = \delta_{s,s'} e_s$ gives the internal direct sum $C = \bigoplus_{s=1}^{N} e_s C$. Multiplication by e_s projects $\mathcal{M}_\lambda$ surjectively onto

$$e_s \mathcal{M}_\lambda = e_s \mathcal{A}[x] \Big/ \langle e_s(x^n - \lambda)\rangle \cong K_s[x]/\langle P_s(x)\rangle,$$

carrying the $\mathcal{A}[x]$-submodule C to a $K_s[x]$-ideal $e_s C$; none of these idempotent steps uses square-freeness of $P_s(x)$ (where each component ring $K_s[x]/\langle P_s(x)\rangle \cong \prod_{f \in \mathcal{F}_s} K_s[x]/\langle f(x)^M\rangle$ is a product of local principal chain rings rather than fields whenever $M > 1$).

Applying Lemma 2.6 to each component ideal $e_s C$ produces a unique $e^{(s)} \in \{0, \ldots, M\}^{\mathcal{F}_s}$ with $e_s C = C_s(e^{(s)})$, proving surjectivity. Conversely, each $C(e)$ in (2.11) is an ideal of $\prod_{s=1}^{N} K_s[x]/\langle P_s(x)\rangle \cong \mathcal{M}_\lambda$, and if $C(e) = C(e')$ then multiplying by e_s yields $C_s(e^{(s)}) = C_s(e'^{(s)})$ and hence $e^{(s)} = e'^{(s)}$ for all s by Lemma 2.6. Thus $e \mapsto C(e)$ is bijective, so $|\mathscr{C}^{(M)}| = \prod_{s=1}^{N} (M+1)^{|\mathcal{F}_s|} = (M+1)^{N_F}$. Since $[K_s : \mathbb{F}_q] = m_s$, summing the $\mathbb{F}_q$-dimensions of the direct summands and substituting (2.9) gives (2.13). □

2.4. Standing assumptions and notation

For ease of reference, we collect our standing conventions in one place:

- **A1.** $\mathcal{A}$ is square-free over $\mathbb{F}_q$ ($q = p^e$, $e \geq 1$) with fixed labeled decomposition $\mathcal{A} \cong \prod_{s=1}^{N} K_s$, $K_s = \mathbb{F}_{q^{m_s}} = \mathbb{F}_{p^{d_s}}$, $d_s = e m_s$; only the constacyclic modulus $P_s(x) = x^n - \lambda_s$ may have repeated roots.
- **A2.** $n = n' p^t \geq 1$ with $\gcd(n', p) = 1$, $t = v_p(n) \geq 0$, and multiplicity $M = p^t$ ($t = 0$, i.e. $M = 1$, is the simple-root case).
- **A3.** $\lambda = (\lambda_s)_{s=1}^{N} \in \mathcal{A}^*$ is an arbitrary unit, with unique component M-th roots $\lambda_{s,0} \in K_s^*$ satisfying $\lambda_{s,0}^M = \lambda_s$. The reciprocal-orbit product is formulated for compatible components (§3.4), while incompatible components are resolved completely by Proposition 3.8.
- **A4.** Codes are $\mathcal{A}$-linear λ-constacyclic codes of length n in the sense of Definition 2.4.
- **A5.** Global dimensions (κ, η_k, W) are taken over $\mathbb{F}_q$; component dimensions over K_s multiply by $m_s = [K_s : \mathbb{F}_q]$.
- **A6.** Counting is labeled: distinct exponent selections $e \neq e'$ in $\mathscr{C}^{(M)}$ count as distinct codes in $\mathcal{A}^n$, without quotienting by monomial, Galois, or ring automorphisms.

A7. The k-Galois pairing ($0 \le k < e$) is code-first, $\langle x, y\rangle_{s,k} = \sum_{i=0}^{n-1} x_i y_i^{p^k}$ (see Remark 3.10 for the candidate-first convention).

A8. Probabilistic expectations, variances, and conditional laws refer to the uniform measure $\mathbb{P}(C(e)) = (M+1)^{-N_F}$ on $\mathscr{C}^{(M)}$.

The symbols used throughout the manuscript are summarised in Table 2; none is overloaded elsewhere in the text.

Table 2: Master notation and dimension conventions used throughout the paper.

Symbol	Meaning
p, $q = p^e$, k	prime characteristic, base-field order ($e \ge 1$), and Galois twist exponent ($0 \le k < e$)
$s \in \{1, \dots, N\}$	fixed component index in $\mathcal{A} \cong \prod_{s=1}^{N} K_s$
$K_s = \mathbb{F}_{q^{m_s}} = \mathbb{F}_{p^{d_s}}$	s-th component field, with $m_s = [K_s : \mathbb{F}_q]$ and $d_s = em_s$
$n = n'p^t$, $M = p^t$	code length with $\gcd(n', p) = 1$, $t = v_p(n) \ge 0$, and factor multiplicity M
λ_s, $\lambda_{s,0} \in K_s^*$	component twist and its unique M-th root satisfying $\lambda_{s,0}^M = \lambda_s$
$P_s(x) = x^n - \lambda_s$	component constacyclic modulus, equal to $(x^{n'} - \lambda_{s,0})^M$
$\mathcal{F}_s$	set of distinct monic irreducible factors of $x^{n'} - \lambda_{s,0}$ over K_s
$e = (e_f)_{f \in \mathcal{F}_s}$, $\mathscr{C}^{(M)}$	exponent selection with $0 \le e_f \le M$, and global selection space $\prod_s \{0, \dots, M\}^{\mathcal{F}_s}$
$g_e(x)$, $h_e(x)$	generator $\prod_{f \in \mathcal{F}_s} f^{e_f}$ and parity-check polynomial $P_s/g_e = \prod_{f \in \mathcal{F}_s} f^{M - e_f}$
$C_s(e)$, $C(e)$	component code $\langle g_e \rangle \subseteq K_s[x]/\langle P_s \rangle$ and global code $\prod_s C_s(e^{(s)}) \subseteq \mathcal{A}^n$
$\sigma_{s,k}$, $\rho_{s,k}$	k-Frobenius $a \mapsto a^{p^k}$ on K_s and its inverse $a \mapsto a^{p^{d_s - k}}$
$f^{\#_{s,k}}$, $\tau_{s,k}$	normalised inverse-Frobenius reciprocal (3.7) and induced permutation of $\mathcal{F}_s$
$\mathcal{O}_s$, O, a_O, d_O, w_O	orbit set of $\tau_{s,k}$, an orbit $O \in \mathcal{O}_s$, length $\lvert O\rvert$, factor degree, and $\mathbb{F}_q$-weight $m_s d_O$
$e_{O,i}$ ($i \in \mathbb{Z}/a_O\mathbb{Z}$)	cyclic exponent word along O, ordered by $\tau_{s,k}(f_{O,i}) = f_{O,i+1}$
$D_k(\cdot)$, $\mathrm{Hull}_k(\cdot)$	component k-Galois dual and hull; globally $D_{\mathcal{A},k}(\cdot)$ and $\mathrm{Hull}_{\mathcal{A},k}(\cdot)$
$\kappa(C)$, $\eta_k(C)$	$\mathbb{F}_q$-dimensions $\dim_{\mathbb{F}_q} C$ and $\dim_{\mathbb{F}_q} \mathrm{Hull}_{\mathcal{A},k}(C)$
N_F, W	total distinct-factor count $\sum_s \lvert\mathcal{F}_s\rvert$ and ambient $\mathbb{F}_q$-dimension $n \sum_s m_s$
U, Z	formal variables marking $\kappa(C)$ and $\eta_k(C)$ in generating polynomials
$\mathscr{E}^{(M)}(U, Z)$, $\mathscr{H}^{(M)}(Z)$	exact joint enumerator and marginal hull polynomial $\mathscr{E}^{(M)}(1, Z)$
$T_w^{(M)}(U, Z)$	$(M+1) \times (M+1)$ transfer matrix on state set $\{0, \dots, M\}$
$\Phi_j(\kappa)$	factorial-moment coefficient $[U^\kappa]\, \partial_Z^j \mathscr{E}^{(M)}(U, Z)\big\rvert_{Z=1}$ ($j = 0, 1, 2$)

3. k-Galois duality

We next determine the k-Galois dual of an arbitrary repeated-root constacyclic code over $\mathcal{A}$ and classify twist compatibility across all units $\lambda \in \mathcal{A}^*$. We recall the componentwise and global k-Galois pairings (§3.1) and the normalised inverse-Frobenius reciprocal (§3.2), recheck the Euclidean dual of a constacyclic ideal directly without any square-freeness assumption and deduce the repeated-root k-Galois dual generator (§3.3), and establish the unchanged compatibility criterion together with the exact behaviour of incompatible twists and dual conventions (§3.4).

3.1. Component and global k-Galois pairings

Fix an integer k with $0 \le k < e$. On each component field $K_s = \mathbb{F}_{q^{m_s}} = \mathbb{F}_{p^{d_s}}$ ($d_s = em_s > k$), define the field automorphisms

$$\sigma_{s,k}(a) := a^{p^k}, \qquad \rho_{s,k}(a) := \sigma_{s,k}^{-1}(a) = a^{p^{d_s-k}} \qquad (a \in K_s). \tag{3.1}$$

For vectors $x = (x_0, \dots, x_{n-1})$ and $y = (y_0, \dots, y_{n-1})$ in K_s^n, the code-first k-Galois pairing [11] is

$$\langle x, y\rangle_{s,k} := \sum_{i=0}^{n-1} x_i \sigma_{s,k}(y_i) = \sum_{i=0}^{n-1} x_i y_i^{p^k}, \tag{3.2}$$

and the component k-Galois dual and k-Galois hull of a K_s-linear code $C \subseteq K_s^n$ are

$$D_k(C) := \{y \in K_s^n : \langle c, y\rangle_{s,k} = 0 \text{ for all } c \in C\}, \qquad \mathrm{Hull}_k(C) := C \cap D_k(C).$$

When $k = 0$, (3.2) is the standard Euclidean inner product on every component K_s^n; we write $C^{\perp_E} := D_0(C)$ for the Euclidean dual. On $K_s = \mathbb{F}_{p^{d_s}}$, the pairing (3.2) is Hermitian if and only if d_s is even and $k = d_s/2$; in particular, when e is even, the choice $k = e/2$ is Hermitian on components with $m_s = 1$ and never Hermitian on components with $m_s > 1$.

Globally on $\mathcal{A} \cong \prod_{s=1}^N K_s$, set $\sigma_{\mathcal{A},k}((a_s)_s) := (\sigma_{s,k}(a_s))_s$ and $\rho_{\mathcal{A},k}((a_s)_s) := (\rho_{s,k}(a_s))_s$, and define the $\mathcal{A}$-valued pairing on $\mathcal{A}^n$ by

$$\langle x, y\rangle_{\mathcal{A},k} := \sum_{i=0}^{n-1} x_i \sigma_{\mathcal{A},k}(y_i) = \Big(\langle x_s, y_s\rangle_{s,k}\Big)_{s=1}^N, \tag{3.3}$$

with global dual and hull

$$D_{\mathcal{A},k}(C) := \{y \in \mathcal{A}^n : \langle c, y\rangle_{\mathcal{A},k} = 0_{\mathcal{A}} \text{ for all } c \in C\}, \qquad \mathrm{Hull}_{\mathcal{A},k}(C) := C \cap D_{\mathcal{A},k}(C).$$

Lemma 3.1 (Dual translation and product decomposition)**.** *Let $n \ge 1$ be arbitrary.*

(a) *For every K_s-linear subspace $C \subseteq K_s^n$, applying $\rho_{s,k}$ coordinatewise gives*

$$D_k(C) = \rho_{s,k}\Big(C^{\perp_E}\Big), \qquad \textit{so} \qquad \dim_{K_s} D_k(C) = n - \dim_{K_s} C. \tag{3.4}$$

(b) *For every global exponent selection $e = (e^{(s)})_{s=1}^N \in \mathscr{C}^{(M)}$,*

$$D_{\mathcal{A},k}\Big(C(e)\Big) = \prod_{s=1}^N D_k\Big(C_s(e^{(s)})\Big), \qquad \mathrm{Hull}_{\mathcal{A},k}\Big(C(e)\Big) = \prod_{s=1}^N \mathrm{Hull}_k\Big(C_s(e^{(s)})\Big), \tag{3.5}$$

and the global $\mathbb{F}_q$-hull dimension is

$$\eta_k\Big(C(e)\Big) := \dim_{\mathbb{F}_q} \mathrm{Hull}_{\mathcal{A},k}\Big(C(e)\Big) = \sum_{s=1}^N m_s \dim_{K_s} \mathrm{Hull}_k\Big(C_s(e^{(s)})\Big). \tag{3.6}$$

Proof. For (a), a vector $y \in K_s^n$ satisfies $\sum_{i=0}^{n-1} c_i \sigma_{s,k}(y_i) = 0$ for all $c \in C$ if and only if $z := \sigma_{s,k}(y) \in C^{\perp_E}$, i.e. $y = \rho_{s,k}(z)$ with $z \in C^{\perp_E}$. Since $\rho_{s,k} : K_s^n \to K_s^n$ is a bijection of finite sets and a $\rho_{s,k}$-semilinear vector-space isomorphism, $\dim_{K_s} D_k(C) = \dim_{K_s} C^{\perp_E} = n - \dim_{K_s} C$.

For (b), testing $\langle c, y\rangle_{\mathcal{A},k} = 0_{\mathcal{A}}$ in (3.3) against vectors $c \in C(e)$ supported solely in coordinate s shows $y_s \in D_k(C_s(e^{(s)}))$ for each s, and the converse inclusion is immediate componentwise. Intersecting with $C(e) = \prod_{s=1}^N C_s(e^{(s)})$ gives the hull decomposition in (3.5), and summing $\mathbb{F}_q$-dimensions with $[K_s : \mathbb{F}_q] = m_s$ gives (3.6). None of these vector-space arguments depends on $\gcd(n, p)$. □

3.2. The inverse-Frobenius reciprocal

For a monic polynomial $f(x) = \sum_{i=0}^{r} f_i x^i \in K_s[x]$ of degree $r \geq 0$ with $f_0 \neq 0$, define its normalised inverse-Frobenius reciprocal by

$$f^{\#_{s,k}}(x) := \rho_{s,k}(f_0)^{-1} \sum_{i=0}^{r} \rho_{s,k}(f_i)\, x^{r-i}. \tag{3.7}$$

Lemma 3.2 (Properties of the twisted reciprocal)**.** *Let $f, g \in K_s[x]$ be monic polynomials with nonzero constant terms.*

(a) *$f^{\#_{s,k}}$ is monic of degree* $\deg f$ *with constant term $\rho_{s,k}(f_0)^{-1} \neq 0$, and the map $f \mapsto f^{\#_{s,k}}$ is a bijection on the set of monic polynomials in $K_s[x]$ with nonzero constant term, inverted by $h(x) \mapsto \sigma_{s,k}(h_0)^{-1} \sum_{i=0}^{\deg h} \sigma_{s,k}(h_i) x^{\deg h - i}$.*

(b) *$(fg)^{\#_{s,k}} = f^{\#_{s,k}} g^{\#_{s,k}}$; in particular,*

$$(f^m)^{\#_{s,k}} = \left(f^{\#_{s,k}}\right)^m \qquad \text{for every integer } m \geq 0. \tag{3.8}$$

(c) *$f^{\#_{s,k}}$ is irreducible in $K_s[x]$ if and only if f is irreducible in $K_s[x]$. Moreover, if L/K_s is a finite extension and $\alpha \in L^*$ satisfies $f(\alpha) = 0$, then $f^{\#_{s,k}}\left(\alpha^{-p^{d_s - k}}\right) = 0$.*

Proof. In (3.7) the coefficient of x^r is $\rho_{s,k}(f_0)^{-1}\rho_{s,k}(f_0) = 1$ and the constant term is $\rho_{s,k}(f_0)^{-1}\rho_{s,k}(1) = \rho_{s,k}(f_0)^{-1} \neq 0$; applying the displayed $\sigma_{s,k}$-reciprocal directly returns f, proving (a). Writing (3.7) in $K_s[x, x^{-1}]$ as

$$f^{\#_{s,k}}(x) = \rho_{s,k}(f_0)^{-1} x^{\deg f}\, (\widetilde{\rho}_{s,k} f)(x^{-1}), \tag{3.9}$$

where $\widetilde{\rho}_{s,k} : K_s[x] \to K_s[x]$ applies $\rho_{s,k}$ coefficientwise, yields $(fg)^{\#_{s,k}} = f^{\#_{s,k}} g^{\#_{s,k}}$ and hence (3.8) by induction on m (with $1^{\#_{s,k}} = 1$ at $m = 0$). By (a) and (b), non-trivial monic factorisations of f correspond bijectively to non-trivial monic factorisations of $f^{\#_{s,k}}$, so irreducibility is preserved. Finally, extend $\alpha \mapsto \alpha^{p^{d_s - k}}$ to L; if $f(\alpha) = 0$, evaluating (3.9) at $\beta = \alpha^{-p^{d_s - k}} \in L^*$ gives

$$\left(\widetilde{\rho}_{s,k} f\right)(\beta^{-1}) = \sum_{i=0}^{r} f_i^{p^{d_s - k}} \left(\alpha^{p^{d_s - k}}\right)^i = \left(f(\alpha)\right)^{p^{d_s - k}} = 0,$$

whence $f^{\#_{s,k}}(\beta) = 0$, establishing (c). □

3.3. Euclidean and k-Galois dual generators

The bridge from Euclidean duality to k-Galois duality in Lemma 3.1(a) requires the generator polynomial of the Euclidean dual of a constacyclic code of arbitrary length n. We verify this directly line by line so it is transparent that repeated roots cause no obstruction.

Lemma 3.3 (Euclidean dual without square-freeness)**.** *Let K be a finite field, $a \in K^*$, and $n \geq 1$ be arbitrary. Let $G(x) \in K[x]$ be a monic divisor of $x^n - a$, set $H(x) := (x^n - a)/G(x)$, and write*

$$H^*(x) := H(0)^{-1} \sum_{v=0}^{\deg H} H_v x^{\deg H - v}$$

for the normalised ordinary reciprocal of $H(x) = \sum_{v=0}^{\deg H} H_v x^v$. Under the coefficient-vector identification $K[x]/\langle x^n - a\rangle \cong K^n$, the Euclidean dual $\langle G(x)\rangle^{\perp_E}$ is the a^{-1}-constacyclic ideal generated by $H^*(x)$ in $K[x]/\langle x^n - a^{-1}\rangle$.

Proof. Put $r := \deg G$ and $h := \deg H$, so $0 \le r, h \le n$, $r + h = n$, and both G and H are monic with $G(0)H(0) = (GH)(0) = -a \neq 0$. Thus $G^*(x) = G(0)^{-1}x^r G(x^{-1})$ and $H^*(x) = H(0)^{-1}x^h H(x^{-1})$ are well-defined monic polynomials of degrees r and h in $K[x]$, and in $K[x, x^{-1}]$ (hence in $K[x]$) we have

$$G^*(x)\,H^*(x) = \Big(G(0)H(0)\Big)^{-1} x^{r+h} G(x^{-1})H(x^{-1}) = (-a)^{-1}x^n(x^{-n} - a) = x^n - a^{-1}. \quad (3.10)$$

In particular, $H^*(x)$ is a monic divisor of $x^n - a^{-1}$, so $\langle H^*(x)\rangle$ is an ideal of $K[x]/\langle x^n - a^{-1}\rangle$ of K-dimension $n - h = r$ spanned by the non-wrapped shifts $\{x^j H^*(x) : 0 \le j < r\}$, while $\langle G(x)\rangle \subseteq K[x]/\langle x^n - a\rangle$ has K-dimension $n - r = h$ spanned by $\{x^i G(x) : 0 \le i < h\}$ (an empty spanning set having span $\{0\}$ when $r = 0$ or $h = 0$).

Write $G(x) = \sum_{u=0}^{r} G_u x^u$. For any $0 \le i < h$ and $0 \le j < r$, every term in $x^i G(x)$ and $x^j H^*(x)$ has degree strictly less than n, so the Euclidean inner product of their coefficient vectors in K^n is

$$\langle x^i G,\, x^j H^*\rangle_{s,0} = H(0)^{-1} \sum_{\substack{0\le u\le r,\ 0\le v\le h \\ i+u=j+h-v}} G_u H_v = H(0)^{-1}[x^{h-i+j}]\Big(G(x)H(x)\Big) = H(0)^{-1}[x^{h-i+j}](x^n - a) = 0,$$

because $0 \le i \le h - 1$ and $0 \le j \le r - 1$ imply $1 \le h - i + j \le h + r - 1 = n - 1$. Hence $\langle H^*(x)\rangle \subseteq \langle G(x)\rangle^{\perp_E}$, and equality follows from $\dim_K \langle H^*(x)\rangle = r = n - h = \dim_K \langle G(x)\rangle^{\perp_E}$. No step uses that $x^n - a$ is square-free; multiplicities are carried along by H. □

Theorem 3.4 (k-Galois dual generator)**.** *Let $e \in \{0, \dots, M\}^{\mathcal{F}_s}$, let $C_s(e) = \langle g_e(x)\rangle \subseteq K_s[x]/\langle P_s(x)\rangle$, and let $h_e(x) = P_s(x)/g_e(x) = \prod_{f\in\mathcal{F}_s} f(x)^{M-e_f}$ be as in (2.8). Then the k-Galois dual $D_k(C_s(e))$ is the $\lambda_s^{-p^{d_s-k}}$-constacyclic ideal in $K_s[x]/\langle x^n - \lambda_s^{-p^{d_s-k}}\rangle$ generated by the monic polynomial*

$$h_e^{\#_{s,k}}(x) = \prod_{f\in\mathcal{F}_s} \Big(f^{\#_{s,k}}(x)\Big)^{M-e_f}. \quad (3.11)$$

If, in addition, $f \mapsto f^{\#_{s,k}}$ restricts to a bijection $\tau_{s,k} : \mathcal{F}_s \to \mathcal{F}_s$, then

$$h_e^{\#_{s,k}}(x) = \prod_{f\in\mathcal{F}_s} f(x)^{\,M-e_{\tau_{s,k}^{-1}(f)}}, \quad (3.12)$$

so the multiplicity of $f \in \mathcal{F}_s$ in the dual generator is $M - e_{\tau_{s,k}^{-1}(f)}$.

Proof. By Lemma 3.3 with $K = K_s$, $a = \lambda_s$, $G = g_e$, and $H = h_e$, the Euclidean dual $C_s(e)^{\perp_E}$ is the λ_s^{-1}-constacyclic ideal $\langle h_e^*(x)\rangle \subseteq K_s[x]/\langle x^n - \lambda_s^{-1}\rangle$. Coefficientwise application of $\rho_{s,k}$ defines a ring automorphism $\widetilde{\rho}_{s,k} : K_s[x] \to K_s[x]$ that sends $x^n - \lambda_s^{-1}$ to $x^n - \rho_{s,k}(\lambda_s^{-1}) = x^n - \lambda_s^{-p^{d_s-k}}$ and sends $h_e^*(x)$ to

$$\widetilde{\rho}_{s,k}\Big(h_e^*(x)\Big) = \rho_{s,k}\Big(h_e(0)\Big)^{-1} \sum_{v=0}^{\deg h_e} \rho_{s,k}(h_{e,v}) x^{\deg h_e - v} = h_e^{\#_{s,k}}(x).$$

Under the coefficient-vector identification $K_s[x]/\langle x^n - \mu\rangle \cong K_s^n$, $\widetilde{\rho}_{s,k}$ acts on K_s^n coordinatewise as $\rho_{s,k}$. Combining this with $D_k(C_s(e)) = \rho_{s,k}(C_s(e)^{\perp_E})$ from Lemma 3.1(a) shows that $D_k(C_s(e))$ is the $\lambda_s^{-p^{d_s-k}}$-constacyclic ideal generated by $h_e^{\#_{s,k}}(x)$.

Applying Lemma 3.2(b) to $h_e = \prod_{f\in\mathcal{F}_s} f^{M-e_f}$ gives (3.11). When $\tau_{s,k}(f) = f^{\#_{s,k}}$ permutes $\mathcal{F}_s$, reindexing the product in (3.11) via $f' = \tau_{s,k}(f)$ (so $f = \tau_{s,k}^{-1}(f')$) yields (3.12). □

3.4. Compatibility, general twists, and dual conventions

We now show that passing from the length-n twist λ_s to its unique M-th root $\lambda_{s,0}$ leaves the simple-root twist-compatibility criterion completely unchanged.

Proposition 3.5 (Compatibility in the repeated-root setting)**.** *For each component $1 \le s \le N$, the following conditions are equivalent:*

(a) $\lambda_{s,0}^{1+p^{d_s-k}} = 1$;
(b) $\lambda_s^{1+p^{d_s-k}} = 1$;
(c) *the dual twist* $\lambda_s^{-p^{d_s-k}}$ *equals* λ_s;
(d) *the map* $f \mapsto f^{\#_{s,k}}$ *maps* $\mathcal{F}_s$ *into* $\mathcal{F}_s$, *and hence restricts to a degree-preserving permutation* $\tau_{s,k} : \mathcal{F}_s \to \mathcal{F}_s$, $\tau_{s,k}(f) := f^{\#_{s,k}}$.

Proof. Put $c := 1 + p^{d_s-k}$. Using $\lambda_s = \lambda_{s,0}^M$ from (2.5),

$$\lambda_s^c = \left(\lambda_{s,0}^M\right)^c = \left(\lambda_{s,0}^c\right)^M.$$

Since $y \mapsto y^M = y^{p^t}$ is an automorphism of K_s^*, we have $(\lambda_{s,0}^c)^M = 1$ if and only if $\lambda_{s,0}^c = 1$, proving (a) $\iff$ (b). Multiplying $\lambda_s^{-p^{d_s-k}} = \lambda_s$ by $\lambda_s^{p^{d_s-k}} \in K_s^*$ gives (b) $\iff$ (c).

Suppose (a) holds and let $f \in \mathcal{F}_s$. Choose a finite extension L/K_s and a root $\alpha \in L^*$ of f, so $\alpha^{n'} = \lambda_{s,0}$. By Lemma 3.2(c), $\beta := \alpha^{-p^{d_s-k}} \in L^*$ is a root of the monic K_s-irreducible polynomial $f^{\#_{s,k}}$, and

$$\beta^{n'} = \left(\alpha^{n'}\right)^{-p^{d_s-k}} = \lambda_{s,0}^{-p^{d_s-k}} = \lambda_{s,0}$$

by (a). Thus $f^{\#_{s,k}}$ shares a root in L with $x^{n'} - \lambda_{s,0} \in K_s[x]$, so $f^{\#_{s,k}} \mid (x^{n'} - \lambda_{s,0})$ in $K_s[x]$, i.e. $f^{\#_{s,k}} \in \mathcal{F}_s$. Because $f \mapsto f^{\#_{s,k}}$ is injective and preserves degrees by Lemma 3.2(a), restriction to the finite set $\mathcal{F}_s$ yields a degree-preserving permutation $\tau_{s,k} : \mathcal{F}_s \to \mathcal{F}_s$, proving (a) $\Longrightarrow$ (d).

Conversely, suppose (d) holds. Since $n' \ge 1$, $\mathcal{F}_s$ is non-empty; choose $f \in \mathcal{F}_s$ and a root $\alpha \in L^*$ of f. Because $f^{\#_{s,k}} \in \mathcal{F}_s$, its root $\alpha^{-p^{d_s-k}}$ also satisfies $(\alpha^{-p^{d_s-k}})^{n'} = \lambda_{s,0}$, whence $\lambda_{s,0}^{-p^{d_s-k}} = \lambda_{s,0}$ and therefore $\lambda_{s,0}^{1+p^{d_s-k}} = 1$. □

Remark 3.6 (Cyclic and negacyclic codes). Both $\lambda_s = 1$ (cyclic codes, with $\lambda_{s,0} = 1$) and $\lambda_s = -1$ (negacyclic codes, with $\lambda_{s,0} = -1$ since $(-1)^{p^t} = -1$ in every characteristic p) satisfy condition (b) of Proposition 3.5 for all p, d_s, k, and n: if $p = 2$ then $-1 = 1$, while if p is odd then $1 + p^{d_s-k}$ is even so $(-1)^{1+p^{d_s-k}} = 1$.

Example 3.7 (Non-cyclic compatible dual generator)**.** Let $q = 4$ ($p = 2$, $e = 2$), $N = 1$, $K_1 = \mathbb{F}_4 = \mathbb{F}_2(\omega)$ with $\omega^2 + \omega + 1 = 0$ ($m_1 = 1$, $d_1 = 2$), $k = 1$, and $n = 10 = 5 \cdot 2^1$ ($n' = 5$, $t = 1$, $M = 2$). Take $\lambda_{1,0} = \omega$, so the constacyclic shift at length $n = 10$ is $\lambda_1 = \lambda_{1,0}^2 = \omega^2 \ne \lambda_{1,0}$. Since $\lambda_{1,0}^{1+2^{2-1}} = \omega^3 = 1$ (equivalently $\lambda_1^3 = \omega^6 = 1$), Proposition 3.5 applies. Over $\mathbb{F}_4$, the base polynomial $x^5 - \omega$ factors into pairwise distinct monic irreducibles as

$$x^5 - \omega = (x + \omega^2)(x^2 + \omega x + \omega)(x^2 + x + \omega) =: f_1(x) f_2(x) f_3(x),$$

so $P_1(x) = x^{10} - \omega^2 = f_1(x)^2 f_2(x)^2 f_3(x)^2$. With $\rho_{1,1}(a) = a^2$ on $\mathbb{F}_4$, (3.7) gives

$$f_1^{\#_{1,1}}(x) = \omega^{-4}(\omega^4 + x) = x + \omega^2 = f_1(x), \qquad f_2^{\#_{1,1}}(x) = \omega^{-2}(\omega^2 + \omega^2 x^1 + x^2) = x^2 + x + \omega = f_3(x),$$

and hence $f_3^{\#_{1,1}}(x) = f_2(x)$, so $\tau_{1,1}$ has orbits (f_1) of length 1 (degree 1) and (f_2, f_3) of length 2 (degree 2). For the exponent selection $e = (e_{f_1}, e_{f_2}, e_{f_3}) = (2, 1, 0)$, Theorem 3.4 gives $g_e = f_1^2 f_2$, parity-check polynomial $h_e = f_2 f_3^2$, and 1-Galois dual generator

$$h_e^{\#_{1,1}} = f_1^{2-e_{f_1}} f_2^{2-e_{f_3}} f_3^{2-e_{f_2}} = f_2^2 f_3,$$

with $\dim_{\mathbb{F}_4} C_1(e) = 10-4 = 6$, $\mathrm{Hull}_1(C_1(e)) = \langle \mathrm{lcm}(g_e, h_e^{\#_{1,1}}) \rangle = \langle f_1^2 f_2^2 f_3 \rangle$, and $\dim_{\mathbb{F}_4} \mathrm{Hull}_1(C_1(e)) = 10 - 8 = 2$.

Components violating Proposition 3.5 require no orbit machinery: every code on such a component has trivial k-Galois hull.

Proposition 3.8 (Incompatible twists and mixed algebras)**.** *Suppose component $s \in \{1, \dots, N\}$ fails compatibility, i.e. $\lambda_{s,0}^{1+p^{d_s-k}} \neq 1$ (equivalently $\mu_s := \lambda_s^{-p^{d_s-k}} \neq \lambda_s$).*

(a) *For every exponent selection $e \in \{0, \dots, M\}^{\mathcal{F}_s}$,*

$$\mathrm{Hull}_k\Big(C_s(e)\Big) = \{0\};$$

thus all $(M+1)^{|\mathcal{F}_s|}$ λ_s-constacyclic codes on component s are k-Galois LCD.

(b) *On component s, the unique k-Galois self-orthogonal ($C \subseteq D_k(C)$) code is $\{0\}$ ($e_f = M$ for all $f \in \mathcal{F}_s$), the unique k-Galois dual-containing ($D_k(C) \subseteq C$) code is K_s^n ($e_f = 0$ for all $f \in \mathcal{F}_s$), and there is no k-Galois self-dual ($C = D_k(C)$) λ_s-constacyclic code.*

(c) *For an arbitrary global unit $\lambda = (\lambda_s)_{s=1}^N \in \mathcal{A}^*$, partition $\{1, \dots, N\} = S_{\mathrm{comp}} \sqcup S_{\mathrm{inc}}$ into compatible and incompatible components. Then*

$$\sum_{e \in \mathscr{C}^{(M)}} U^{\kappa(C(e))} Z^{\eta_k(C(e))} = \left(\prod_{s \in S_{\mathrm{comp}}} \sum_{e^{(s)} \in \{0,\dots,M\}^{\mathcal{F}_s}} U^{m_s \dim_{K_s} C_s(e^{(s)})} Z^{m_s \dim_{K_s} \mathrm{Hull}_k(C_s(e^{(s)}))} \right) \prod_{s \in S_{\mathrm{inc}}} \prod_{f \in \mathcal{F}_s} \left(\sum_{r=0}^{M} U \right. \tag{3.13}$$

Proof. For (a), Theorem 3.4 gives $g_e \mid (x^n - \lambda_s)$ and $h_e^{\#_{s,k}} \mid (x^n - \mu_s)$ in $K_s[x]$, with $\mu_s \neq \lambda_s$ by Proposition 3.5(c). Since $(x^n - \lambda_s) - (x^n - \mu_s) = \mu_s - \lambda_s \in K_s^*$, the polynomials $x^n - \lambda_s$ and $x^n - \mu_s$ are coprime in $K_s[x]$, and hence so are their divisors g_e and $h_e^{\#_{s,k}}$. Let $v = (v_0, \dots, v_{n-1}) \in C_s(e) \cap D_k(C_s(e))$ and put $v(x) := \sum_{i=0}^{n-1} v_i x^i \in K_s[x]$. Because $v \in C_s(e) = \langle g_e \rangle \subseteq K_s[x]/\langle x^n - \lambda_s \rangle$ and $g_e \mid (x^n - \lambda_s)$, we have $v(x) \in \langle g_e,\, x^n - \lambda_s \rangle = \langle g_e \rangle$ in $K_s[x]$, so $g_e \mid v(x)$ in $K_s[x]$. For the same reason, $v \in D_k(C_s(e)) = \langle h_e^{\#_{s,k}} \rangle \subseteq K_s[x]/\langle x^n - \mu_s \rangle$ and $h_e^{\#_{s,k}} \mid (x^n - \mu_s)$ imply $h_e^{\#_{s,k}} \mid v(x)$ in $K_s[x]$. Coprimality in $K_s[x]$ gives $g_e h_e^{\#_{s,k}} \mid v(x)$, whereas $\deg(g_e h_e^{\#_{s,k}}) = \deg g_e + \deg h_e = n > \deg v(x)$; hence $v(x) = 0$.

For (b), if $C \subseteq D_k(C)$ then $C = C \cap D_k(C) = \mathrm{Hull}_k(C) = \{0\}$ by (a); if $D_k(C) \subseteq C$ then $D_k(C) = C \cap D_k(C) = \{0\}$, so $\dim_{K_s} C = n - \dim_{K_s} D_k(C) = n$ by Lemma 3.1(a) and therefore $C = K_s^n$; if $C = D_k(C)$ both hold, forcing $\{0\} = K_s^n$, impossible since $n \geq 1$.

For (c), combine Lemma 2.7, Lemma 3.1(b), and (a): on each $s \in S_{\mathrm{inc}}$ every selection has $\mathrm{Hull}_k(C_s(e^{(s)})) = \{0\}$ and contributes $\sum_{f \in \mathcal{F}_s} (M - e_f) m_s \deg f$ to $\kappa(C(e))$, and setting $r_f = M - e_f \in \{0, \dots, M\}$ factors the sum over $e^{(s)}$ into the product over $f \in \mathcal{F}_s$ displayed in (3.13). □

Example 3.9 (Incompatible repeated-root twist)**.** Take $q = 5$ ($p = 5$, $e = 1$), $N = 1$, $K_1 = \mathbb{F}_5$, $k = 0$, and $n = 10 = 2 \cdot 5^1$ ($n' = 2$, $t = 1$, $M = 5$), with $\lambda_{1,0} = 2$ and $\lambda_1 = \lambda_{1,0}^5 = 2 \in \mathbb{F}_5^*$. Here $\lambda_{1,0}^{1+5^{1-0}} = 2^6 \equiv 4 \not\equiv 1 \pmod 5$, so compatibility fails and $\mu_{1,0} = \mu_1 = 2^{-1} = 3 \in \mathbb{F}_5^*$. Because 2 is a non-square in $\mathbb{F}_5$, $x^2 - 2$ is irreducible over $\mathbb{F}_5$ and $P_1(x) = x^{10} - 2 = (x^2 - 2)^5$, whereas the Euclidean dual modulus is $x^{10} - 3 = (x^2 - 3)^5$. For each exponent $e \in \{0, 1, 2, 3, 4, 5\}$, the code $C_1(e) = \langle (x^2 - 2)^e \rangle$ has Euclidean dual $D_0(C_1(e)) = \langle (x^2 - 3)^{5-e} \rangle$ and exact parameters

$$(\kappa, \eta_0) = (10, 0),\ (8, 0),\ (6, 0),\ (4, 0),\ (2, 0),\ (0, 0),$$

with enumerator $(1 + U^2 + U^4 + U^6 + U^8 + U^{10})Z^0$, unique self-orthogonal code $\{0\}$ ($e = 5$), unique dual-containing code $\mathbb{F}_5^{10}$ ($e = 0$), and no self-dual code, in complete agreement with Proposition 3.8.

Remark 3.10 (Code-first versus candidate-first dual conventions). Some authors (for instance [19, 27]) define the k-Galois dual of $C \subseteq K_s^n$ by the candidate-first condition

$$D_{\mathrm{cand}}(C) := \Big\{ y \in K_s^n : \sum_{i=0}^{n-1} y_i\, \sigma_{s,k}(c_i) = 0 \text{ for all } c \in C \Big\}.$$

For arbitrary $n \geq 1$ and arbitrary K_s-linear subspaces $C \subseteq K_s^n$, substituting $y = \sigma_{s,k}(z)$ gives $D_{\mathrm{cand}}(C) = \sigma_{s,k}(C^{\perp_E}) = \sigma_{s,k}^2(D_k(C))$. Although $D_{\mathrm{cand}}(C)$ and $D_k(C)$ are generally distinct subspaces of K_s^n, if B is any $(\dim_{K_s} C) \times n$ generator matrix of C over K_s and $G := B\, \sigma_{s,k}(B)^{\mathsf{T}}$, then a codeword vB lies in $D_k(C)$ if and only if $G\, \sigma_{s,k}(v)^{\mathsf{T}} = 0$ and lies in $D_{\mathrm{cand}}(C)$ if and only if $vG = 0$. Both nullspaces have dimension $\dim_{K_s} C - \mathrm{rank}(G)$, so

$$\dim_{K_s}\Big(C \cap D_k(C)\Big) = \dim_{K_s} C - \mathrm{rank}\Big(B\, \sigma_{s,k}(B)^{\mathsf{T}}\Big) = \dim_{K_s}\Big(C \cap D_{\mathrm{cand}}(C)\Big). \tag{3.14}$$

Consequently: (i) every dimension statement and every enumerative quantity in this paper — including $\mathscr{E}^{(M)}(U, Z)$, $\mathscr{H}^{(M)}(Z)$, all moments, and the counts of LCD ($\eta_k = 0$), self-orthogonal ($\eta_k = \kappa$), dual-containing ($\eta_k = W - \kappa$), and self-dual ($\eta_k = \kappa = W/2$) codes — is invariant between the code-first and candidate-first conventions; (ii) explicit dual and hull generator polynomials in Theorem 3.4 and Section 4 refer to the code-first pairing (3.2); and (iii) the Gram-rank identity (3.14) gives an independent linear-algebraic test of the hull formulas requiring no factorisation or orbit input (Section 6).

4. Hull statistic and the joint enumerator

Having established the repeated-root k-Galois dual generator in Theorem 3.4 and resolved incompatible components completely in Proposition 3.8, we now express $\mathrm{Hull}_k(C_s(e))$ on every compatible component as an explicit cyclic sum of edge minima along each $\tau_{s,k}$-orbit (§4.1) and lift this local statistic to an exact $(M + 1) \times (M + 1)$ transfer-matrix trace product for the bivariate enumerator $\mathscr{E}^{(M)}(U, Z)$ (§4.2).

4.1. Hull generators, cyclic edge minima, and orbit factorisation

Lemma 4.1 (Intersection of principal ideals modulo P_s)**.** *Let $A(x)$ and $B(x)$ be monic divisors of $P_s(x) = x^n - \lambda_s$ in $K_s[x]$. In the quotient ring $R_s := K_s[x]/\langle P_s(x) \rangle$, we have*

$$\langle A(x) \rangle \cap \langle B(x) \rangle = \Big\langle \mathrm{lcm}\Big(A(x), B(x)\Big) \Big\rangle,$$

where $\operatorname{lcm}(A,B) \in K_s[x]$ *is the unique monic least common multiple of* A *and* B*, and* $\operatorname{lcm}(A,B)$ *divides* $P_s(x)$ *in* $K_s[x]$*.*

Proof. Since P_s is a common multiple of A and B in the principal ideal domain $K_s[x]$, $\operatorname{lcm}(A,B)$ divides P_s. Because $A \mid \operatorname{lcm}(A,B)$ and $B \mid \operatorname{lcm}(A,B)$, we have $\langle \operatorname{lcm}(A,B)\rangle \subseteq \langle A\rangle \cap \langle B\rangle$ in R_s. Conversely, let $u(x)+\langle P_s\rangle \in \langle A\rangle \cap \langle B\rangle$ with $u(x) \in K_s[x]$. Then $u = Aq_1 + P_s r_1$ and $u = Bq_2 + P_s r_2$ for some $q_1, q_2, r_1, r_2 \in K_s[x]$. Since $A \mid P_s$ and $B \mid P_s$, both A and B divide $u(x)$ in $K_s[x]$, so $\operatorname{lcm}(A,B) \mid u(x)$ in $K_s[x]$ and hence $u(x)+\langle P_s\rangle \in \langle \operatorname{lcm}(A,B)\rangle$. □

Now suppose component $s \in \{1,\dots,N\}$ satisfies the compatibility condition of Proposition 3.5, so that $\tau_{s,k}(f) = f^{\#_{s,k}}$ is a degree-preserving permutation of $\mathcal{F}_s$. Decompose $\mathcal{F}_s$ into disjoint cycles (orbits) of $\tau_{s,k}$,

$$\mathcal{F}_s = \bigsqcup_{O \in \mathcal{O}_s} O,$$

and for each orbit $O \in \mathcal{O}_s$ write $a_O := |O| \geq 1$ for its length, $d_O := \deg f$ $(f \in O)$ for the common K_s-degree of its factors, and $w_O := m_s d_O$ for the associated $\mathbb{F}_q$-block weight. Fix a base point $f_{O,0} \in O$ and index the elements of O cyclically by $f_{O,i} := \tau_{s,k}^{i}(f_{O,0})$ for $i \in \mathbb{Z}/a_O\mathbb{Z}$, writing $e_{O,i} := e_{f_{O,i}} \in \{0,1,\dots,M\}$ for the restriction of an exponent selection $e \in \{0,\dots,M\}^{\mathcal{F}_s}$ to O. Note that rechoosing the base point $f_{O,0} \in O$ shifts the cyclic word $(e_{O,0},\dots,e_{O,a_O-1})$ rotationally without changing any cyclic sum below.

Theorem 4.2 (Single-code repeated-root hull generator and dimension)**.** *Suppose component* $s \in \{1,\dots,N\}$ *satisfies* $\lambda_{s,0}^{1+p^{d_s-k}} = 1$*. For every exponent selection* $e \in \{0,\dots,M\}^{\mathcal{F}_s}$*:*

(a) *The* k*-Galois hull* $\operatorname{Hull}_k(C_s(e)) \subseteq K_s[x]/\langle P_s(x)\rangle$ *is the ideal generated by the monic divisor*

$$g_{\mathrm{Hull}}(e;x) := \prod_{f \in \mathcal{F}_s} f(x)^{\max\{e_f,\, M - e_{\tau_{s,k}^{-1}(f)}\}} = \prod_{O \in \mathcal{O}_s} \prod_{i=0}^{a_O-1} f_{O,i}(x)^{\max\{e_{O,i},\, M - e_{O,i-1}\}}. \tag{4.1}$$

(b) *The* K_s*-dimension of* $\operatorname{Hull}_k(C_s(e))$ *is*

$$\dim_{K_s} \operatorname{Hull}_k\Big(C_s(e)\Big) = \sum_{O \in \mathcal{O}_s} d_O \sum_{i=0}^{a_O-1} \min\Big\{e_{O,i},\, M - e_{O,i+1}\Big\}. \tag{4.2}$$

(c) *Globally, for any* $\lambda \in \mathcal{A}^*$ *with partition* $\{1,\dots,N\} = S_{\mathrm{comp}} \sqcup S_{\mathrm{inc}}$ *and any* $e = (e^{(s)})_{s=1}^{N} \in \mathscr{C}^{(M)}$*,*

$$\kappa\Big(C(e)\Big) = \sum_{s \in S_{\mathrm{comp}}} \sum_{O \in \mathcal{O}_s} w_O \sum_{i=0}^{a_O-1} (M - e_{O,i}) + \sum_{s \in S_{\mathrm{inc}}} \sum_{f \in \mathcal{F}_s} m_s(\deg f)(M - e_f), \tag{4.3}$$

and

$$\eta_k\Big(C(e)\Big) = \sum_{s \in S_{\mathrm{comp}}} \sum_{O \in \mathcal{O}_s} w_O \sum_{i=0}^{a_O-1} \min\Big\{e_{O,i},\, M - e_{O,i+1}\Big\}. \tag{4.4}$$

Proof. By Proposition 3.5(c), the dual modulus $x^n - \lambda_s^{-p^{d_s-k}}$ equals $P_s(x) = x^n - \lambda_s$, so both $C_s(e) = \langle g_e \rangle$ and $D_k(C_s(e)) = \langle h_e^{\#_{s,k}} \rangle$ are ideals of the same ring $K_s[x]/\langle P_s(x) \rangle$. Combining (2.8), (3.12), Lemma 4.1, and unique factorisation of pairwise distinct monic irreducibles $f \in \mathcal{F}_s$ gives

$$\operatorname{lcm}\Big(g_e,\, h_e^{\#_{s,k}}\Big) = \prod_{f \in \mathcal{F}_s} f(x)^{\max\{e_f,\, M - e_{\tau_{s,k}^{-1}(f)}\}},$$

which rearranges orbit by orbit (since $\tau_{s,k}^{-1}(f_{O,i}) = f_{O,i-1}$) into (4.1).

Applying (2.9) to the monic divisor $g_{\mathrm{Hull}}(e;x) \mid P_s(x)$ gives

$$\dim_{K_s} \mathrm{Hull}_k\Big(C_s(e)\Big) = \sum_{O \in \mathcal{O}_s} d_O \sum_{i=0}^{a_O-1} \Big(M - \max\Big\{e_{O,i},\, M - e_{O,i-1}\Big\}\Big).$$

For all integers A, B, M one has $M - \max\{A, M-B\} = \min\{M-A, B\}$; applying this identity with $A = e_{O,i}$ and $B = e_{O,i-1}$ and shifting the cyclic index via $j = i - 1 \in \mathbb{Z}/a_O\mathbb{Z}$ yields

$$\sum_{i=0}^{a_O-1} \Big(M - \max\Big\{e_{O,i},\, M - e_{O,i-1}\Big\}\Big) = \sum_{i=0}^{a_O-1} \min\Big\{M - e_{O,i},\, e_{O,i-1}\Big\} = \sum_{j=0}^{a_O-1} \min\Big\{e_{O,j},\, M - e_{O,j+1}\Big\}, \tag{4.5}$$

establishing (4.2). Part (c) follows from (2.13), (3.6), Proposition 3.8(a), and (4.2). □

Remark 4.3 (Reconciliation with Debnath et al. [27, Theorem 8]). In the special case where every component is split over $\mathbb{F}_q$ ($m_s = 1$), $\gcd(\mathrm{ord}(\lambda_s), n') = 1$, and every component is compatible ($\mathrm{ord}(\lambda_s) \mid (1 + p^{e-k})$), the unshifted identity $M - \max\{e_{O,i}, M - e_{O,i-1}\} = \min\{e_{O,i-1}, M - e_{O,i}\}$ in (4.5) agrees term for term with [27, Theorem 8] (see also Remark 3.10 regarding their candidate-first pairing). Writing each summand in the cyclic nearest-neighbour form $\min\{e_{O,i}, M - e_{O,i+1}\}$ has two decisive advantages for enumeration: first, it holds without any restriction on $m_s \geq 1$ or $\gcd(\mathrm{ord}(\lambda_s), n')$; second, each term depends only on a directed edge $(e_{O,i}, e_{O,i+1})$ along the cycle $\mathbb{Z}/a_O\mathbb{Z}$, enabling exact transfer-matrix summation over all $(M+1)^{N_F}$ codes simultaneously.

4.2. Transfer matrices and the exact joint enumerator

For any unit $\lambda \in \mathcal{A}^*$ and $0 \leq k < e$, define the exact bivariate code-dimension and k-Galois-hull enumerator and its marginal hull polynomial in $\mathbb{Z}[U, Z]$ and $\mathbb{Z}[Z]$ by

$$\mathscr{E}^{(M)}(U, Z) := \sum_{e \in \mathscr{C}^{(M)}} U^{\kappa(C(e))}\, Z^{\eta_k(C(e))}, \qquad \mathscr{H}^{(M)}(Z) := \mathscr{E}^{(M)}(1, Z) = \sum_{e \in \mathscr{C}^{(M)}} Z^{\eta_k(C(e))}. \tag{4.6}$$

Because $\mathscr{C}^{(M)} = \prod_{s \in S_{\mathrm{comp}}} \prod_{O \in \mathcal{O}_s} \{0, \ldots, M\}^O \times \prod_{s \in S_{\mathrm{inc}}} \{0, \ldots, M\}^{\mathcal{F}_s}$ is a Cartesian product on which both $\kappa(C(e))$ and $\eta_k(C(e))$ in (4.3)–(4.4) decompose additively with zero cross-terms, the generating sum (4.6) factors over all orbits and incompatible factors into independent local sums.

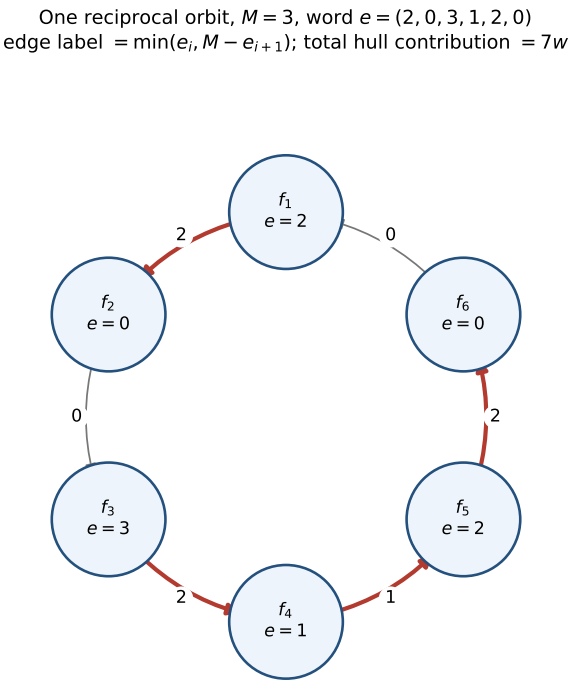


Figure 2: Local cyclic edge-minimum mechanism along a single $\tau_{s,k}$-orbit of length $a = 6$ at multiplicity $M = 3$ for the exponent word $e = (2, 0, 3, 1, 2, 0)$: each directed edge $f_i \to f_{i+1}$ contributes $w \min\{e_i, M - e_{i+1}\}$, giving total orbit hull contribution $7w$.

Definition 4.4 (Repeated-root transfer matrix)**.** For an integer multiplicity $M \geq 1$ and weight $w \geq 1$, the *repeated-root transfer matrix* $T_w^{(M)}(U,Z) \in \mathbb{Z}[U,Z]^{(M+1)\times(M+1)}$ has rows and columns indexed by $\alpha, \beta \in \{0,1,\ldots,M\}$ in increasing order and entries

$$\Big(T_w^{(M)}(U,Z)\Big)_{\alpha,\beta} := U^{w(M-\beta)}\, Z^{w\min\{\alpha,\, M-\beta\}} \qquad (0 \leq \alpha, \beta \leq M), \tag{4.7}$$

so that $T_w^{(M)}(U,Z) = T_1^{(M)}(U^w, Z^w)$. For $M = 1$ and $M = 2$ at unit weight $w = 1$, (4.7) reads explicitly

$$T_1^{(1)}(U,Z) = \begin{pmatrix} U & 1 \\ UZ & 1 \end{pmatrix}, \qquad T_1^{(2)}(U,Z) = \begin{pmatrix} U^2 & U & 1 \\ U^2Z & UZ & 1 \\ U^2Z^2 & UZ & 1 \end{pmatrix}. \tag{4.8}$$

Figure 3 displays the state-transition entries of $T_w^{(2)}(U,Z)$ coloured by the edge hull exponent $\min\{\alpha, M-\beta\}$.

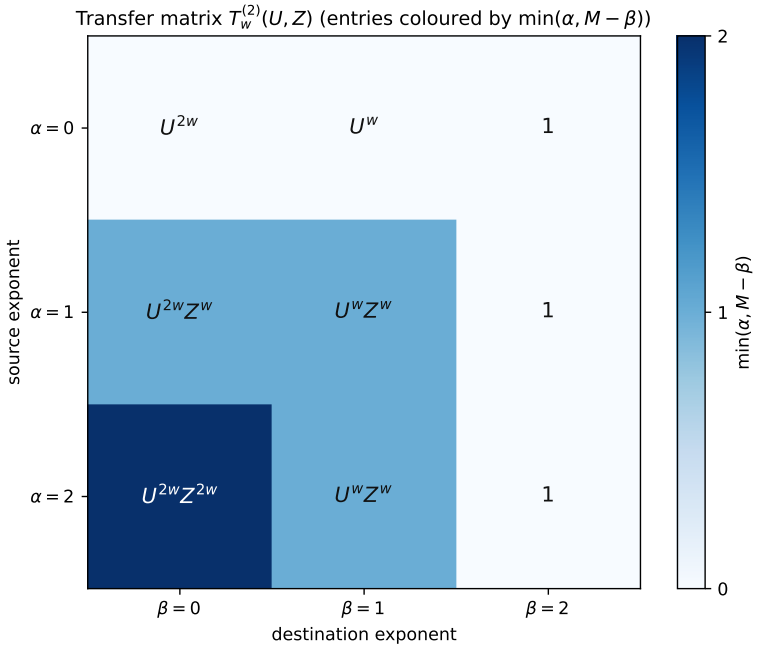


Figure 3: Structure of the $(M+1)\times(M+1)$ repeated-root transfer matrix $T_w^{(2)}(U,Z)$ on states $\alpha, \beta \in \{0,1,2\}$, shaded according to the local hull exponent $\min\{\alpha, 2-\beta\} \in \{0,1,2\}$.

Lemma 4.5 (Single-orbit trace formula)**.** *For every integer $M \geq 1$, orbit length $a \geq 1$, and weight $w \geq 1$, the single-orbit bivariate enumerator*

$$E_{a,w}^{(M)}(U,Z) := \sum_{(e_0,\ldots,e_{a-1}) \in \{0,\ldots,M\}^a} U^{w\sum_{i=0}^{a-1}(M-e_i)}\, Z^{w\sum_{i=0}^{a-1}\min\{e_i,\, M-e_{i+1}\}}$$

(with indices taken modulo a) satisfies

$$E_{a,w}^{(M)}(U,Z) = \mathrm{tr}\Big(T_w^{(M)}(U,Z)^a\Big). \tag{4.9}$$

Proof. Expanding the a-fold matrix product in $\mathbb{Z}[U,Z]^{(M+1)\times(M+1)}$ and taking the trace over closed walks $(e_0, e_1, \dots, e_{a-1}) \in \{0,\dots,M\}^a$ (with $e_a := e_0$) gives

$$\begin{aligned}\operatorname{tr}\Big(T_w^{(M)}(U,Z)^a\Big) &= \sum_{e_0=0}^{M}\sum_{e_1=0}^{M}\cdots\sum_{e_{a-1}=0}^{M}\prod_{i=0}^{a-1}\Big(T_w^{(M)}(U,Z)\Big)_{e_i,e_{i+1}} \\ &= \sum_{(e_0,\dots,e_{a-1})\in\{0,\dots,M\}^a} U^{w\sum_{i=0}^{a-1}(M-e_{i+1})}\, Z^{w\sum_{i=0}^{a-1}\min\{e_i,\,M-e_{i+1}\}}.\end{aligned}$$

Since $\sum_{i=0}^{a-1}(M-e_{i+1}) = \sum_{i=0}^{a-1}(M-e_i)$ around the closed cycle $e_a = e_0$, this coincides identically with $E_{a,w}^{(M)}(U,Z)$. □

Theorem 4.6 (Exact joint code-dimension and k-Galois-hull enumerator)**.** *Let $\mathcal{A} \cong \prod_{s=1}^{N} K_s$ be a finite commutative square-free affine algebra over $\mathbb{F}_q$ ($q = p^e$, $K_s = \mathbb{F}_{q^{m_s}} = \mathbb{F}_{p^{d_s}}$), let $0 \le k < e$, let $n = n'p^t \ge 1$ with $\gcd(n',p) = 1$ and $M = p^t$, and let $\lambda = (\lambda_s)_{s=1}^{N} \in \mathcal{A}^*$ with M-th roots $\lambda_{s,0} \in K_s^*$. Partition $\{1,\dots,N\} = S_{\mathrm{comp}} \sqcup S_{\mathrm{inc}}$ into components satisfying $\lambda_{s,0}^{1+p^{d_s-k}} = 1$ and components failing it. Then:*

(a) *The exact bivariate enumerator* (4.6) *of all $(M+1)^{N_F}$ labeled λ-constacyclic codes of length n over $\mathcal{A}$ is*

$$\mathscr{E}^{(M)}(U,Z) = \left(\prod_{s\in S_{\mathrm{comp}}}\prod_{O\in\mathcal{O}_s}\operatorname{tr}\Big(T_{w_O}^{(M)}(U,Z)^{a_O}\Big)\right)\left(\prod_{s\in S_{\mathrm{inc}}}\prod_{f\in\mathcal{F}_s}\frac{1-U^{(M+1)m_s\deg f}}{1-U^{m_s\deg f}}\right), \quad (4.10)$$

where $a_O = |O|$ and $w_O = m_s \deg f_{O,0}$ for each $\tau_{s,k}$-orbit $O \in \mathcal{O}_s$.

(b) *When every component is compatible ($S_{\mathrm{inc}} = \emptyset$, in particular for all cyclic and negacyclic codes),* (4.10) *simplifies to the pure trace product*

$$\mathscr{E}^{(M)}(U,Z) = \prod_{s=1}^{N}\prod_{O\in\mathcal{O}_s}\operatorname{tr}\Big(T_{w_O}^{(M)}(U,Z)^{a_O}\Big), \qquad \mathscr{H}^{(M)}(Z) = \prod_{s=1}^{N}\prod_{O\in\mathcal{O}_s}\operatorname{tr}\Big(T_{w_O}^{(M)}(1,Z)^{a_O}\Big). \quad (4.11)$$

Proof. Start from Proposition 3.8(c), equation (3.13). On each incompatible component $s \in S_{\mathrm{inc}}$ and each $f \in \mathcal{F}_s$, summing the finite geometric series $\sum_{r=0}^{M} U^{r m_s \deg f} = (1-U^{(M+1)m_s\deg f})/(1-U^{m_s\deg f})$ yields the second factor in (4.10). On each compatible component $s \in S_{\mathrm{comp}}$, identify $\{0,\dots,M\}^{\mathcal{F}_s}$ with $\prod_{O\in\mathcal{O}_s}\{0,\dots,M\}^{a_O}$ via $e^{(s)} \mapsto \Big((e_{O,0},\dots,e_{O,a_O-1})\Big)_{O\in\mathcal{O}_s}$. Substituting (2.9) and (4.2) factors the component sum as

$$\sum_{e^{(s)}\in\{0,\dots,M\}^{\mathcal{F}_s}} U^{m_s\dim_{K_s} C_s(e^{(s)})}\, Z^{m_s\dim_{K_s}\operatorname{Hull}_k(C_s(e^{(s)}))} = \prod_{O\in\mathcal{O}_s} E_{a_O,w_O}^{(M)}(U,Z) = \prod_{O\in\mathcal{O}_s}\operatorname{tr}\Big(T_{w_O}^{(M)}(U,Z)^{a_O}\Big),$$

where the last equality is Lemma 4.5. Specialising $S_{\mathrm{inc}} = \emptyset$ and $U = 1$ gives (4.11). □

Corollary 4.7 (Short orbits and recovery of the simple-root enumerator)**.** *Let $M \geq 1$ and $w \geq 1$.*

(a) (Self-reciprocal orbits, $a = 1$.)

$$\operatorname{tr}\Big(T_w^{(M)}(U,Z)\Big) = \sum_{\alpha=0}^{M} U^{w(M-\alpha)}\, Z^{w\min\{\alpha,\,M-\alpha\}}. \tag{4.12}$$

(b) (Reciprocal pairs, $a = 2$.)

$$\operatorname{tr}\Big(T_w^{(M)}(U,Z)^2\Big) = \sum_{\alpha=0}^{M}\sum_{\beta=0}^{M} U^{w(2M-\alpha-\beta)}\, Z^{w\left(\min\{\alpha,M-\beta\}+\min\{\beta,M-\alpha\}\right)}. \tag{4.13}$$

In particular, writing $x_0 := M-\alpha$ and $x_1 := M-\beta$, every pair with $x_0+x_1 \leq M$ satisfies $\min\{\alpha,M-\beta\}+\min\{\beta,M-\alpha\} = x_0+x_1$ (self-orthogonal), while every pair with $x_0+x_1 \geq M$ satisfies $\min\{\alpha,M-\beta\}+\min\{\beta,M-\alpha\} = 2M-x_0-x_1$ (dual-containing); thus on orbits of length $a \in \{1,2\}$ every code attains the upper boundary $\eta = \min\{\kappa, awM-\kappa\}$.

(c) (Simple-root specialisation, $M = 1$.) *When $t = 0$ ($\gcd(n,p) = 1$, $M = 1$), the matrix $T_w^{(1)}(U,Z) = \left(\begin{smallmatrix} U^w & 1 \\ U^w Z^w & 1 \end{smallmatrix}\right)$ has trace $1+U^w$ and determinant $U^w(1-Z^w)$, and for every $a \geq 1$,*

$$\operatorname{tr}\Big(T_w^{(1)}(U,Z)^a\Big) = \begin{cases} 1+U^w, & a = 1, \\ 1+U^{aw} + \displaystyle\sum_{r=1}^{\lfloor a/2\rfloor}\sum_{c=r}^{a-r} \frac{a}{r}\binom{c-1}{r-1}\binom{a-c-1}{r-1} U^{wc}Z^{wr}, & a \geq 2. \end{cases} \tag{4.14}$$

Proof. Parts (a) and (b) read off the diagonal sums of $T_w^{(M)}(U,Z)$ and $T_w^{(M)}(U,Z)^2$ directly from (4.7); if $x_0+x_1 \leq M$ then $\alpha = M-x_0 \geq x_1 = M-\beta$ and $\beta = M-x_1 \geq x_0 = M-\alpha$, giving $\min\{\alpha,M-\beta\}+\min\{\beta,M-\alpha\} = x_1+x_0$, and similarly for $x_0+x_1 \geq M$. For (c), setting $M = 1$ and $x_i := 1-e_i \in \{0,1\}$ in Lemma 4.5 turns $\min\{e_i, 1-e_{i+1}\} = \min\{1-x_i, x_{i+1}\}$ into the indicator of a cyclic $0 \to 1$ transition $x_i = 0$, $x_{i+1} = 1$. Around a cycle of length a, the number of $0 \to 1$ transitions equals 0 when x is constant ($x = 0^a$ or $x = 1^a$, contributing $1+U^{aw}$) and equals the number $r \in \{1,\dots,\lfloor a/2\rfloor\}$ of maximal circular runs of 1s otherwise. Counting circular binary words of length a with c ones arranged into r circular runs gives $\frac{a}{r}\binom{c-1}{r-1}\binom{a-c-1}{r-1}$ [37], establishing (4.14). □

Example 4.8 (Full bivariate enumerator for Example 3.7)**.** Return to Example 3.7: $K_1 = \mathbb{F}_4 = \mathbb{F}_2(\omega)$, $k = 1$, $n = 10 = 5\cdot 2^1$ ($n' = 5$, $M = 2$), $\lambda_{1,0} = \omega$, $\lambda_1 = \omega^2$, where $\tau_{1,1}$ has one self-reciprocal orbit $O_1 = (f_1)$ with $(a_1,w_1) = (1,1)$ and one reciprocal pair $O_2 = (f_2,f_3)$ with $(a_2,w_2) = (2,2)$. By Corollary 4.7(a)–(b),

$$\operatorname{tr}\Big(T_1^{(2)}(U,Z)\Big) = 1+UZ+U^2, \qquad \operatorname{tr}\Big(T_2^{(2)}(U,Z)^2\Big) = 1+2U^2Z^2+3U^4Z^4+2U^6Z^2+U^8.$$

Multiplying these two local traces as in Theorem 4.6(b) gives all $3^3 = 27$ constacyclic codes in 15 distinct (κ,η_1) classes:

$$\begin{aligned}\mathscr{E}^{(2)}(U,Z) = {} & 1+UZ+U^2+2U^2Z^2+2U^3Z^3+2U^4Z^2+3U^4Z^4+3U^5Z^5 \\ & +2U^6Z^2+3U^6Z^4+2U^7Z^3+U^8+2U^8Z^2+U^9Z+U^{10}.\end{aligned}$$

Setting $Z = 0$ leaves $1 + U^2 + U^8 + U^{10}$ (4 1-Galois LCD codes, equal to $2^{|\mathcal{O}_1|} = 2^2$), while the central term $3U^5Z^5$ counts three 1-Galois self-dual codes arising from the midpoint exponent $e_{f_1} = M/2 = 1$ on O_1 paired with the three solutions of $e_{f_2} + e_{f_3} = 2$ on O_2 — a strictly repeated-root phenomenon absent at simple-root length $n' = 5$, where odd ambient dimension precludes self-dual codes altogether.

5. Consequences

Theorem 4.6 reduces global questions across all $(M+1)^{N_F}$ repeated-root constacyclic codes to explicit algebraic operations on the local $(M+1) \times (M+1)$ transfer matrices $T_w^{(M)}(U, Z)$. We derive in turn the univariate marginals and global duality symmetry (§5.1), the multiplicity invariance of k-Galois LCD codes (§5.2), the exact single-orbit support trapezoid and attainable hull spectrum (§5.3), exact unconditional and conditional moments (§5.4), and closed enumerators for self-orthogonal, dual-containing, and self-dual codes (§5.5).

5.1. Marginals and global duality symmetry

Proposition 5.1 (Marginals and duality reflection)**.** *Let $\lambda \in \mathcal{A}^*$ be any unit, $0 \le k < e$, $n = n'p^t$, $M = p^t$, and $W = n\sum_{s=1}^{N} m_s$.*

(a) *Setting $Z = 1$ eliminates all k-dependence: every row of $T_w^{(M)}(U, 1)$ equals $(U^{wM}, U^{w(M-1)}, \dots, U^w, 1)$, a rank-1 matrix of trace $\sum_{r=0}^{M} U^{rw}$, and*

$$\mathscr{E}^{(M)}(U, 1) = \prod_{s=1}^{N} \prod_{f \in \mathcal{F}_s} \frac{1 - U^{(M+1)m_s \deg f}}{1 - U^{m_s \deg f}}, \qquad \mathscr{E}^{(M)}(1, 1) = \mathscr{H}^{(M)}(1) = (M+1)^{N_F}. \tag{5.1}$$

(b) *The bivariate enumerator satisfies the global functional equation*

$$U^W \, \mathscr{E}^{(M)}\Big(U^{-1}, Z\Big) = \mathscr{E}^{(M)}(U, Z), \tag{5.2}$$

so $[U^\kappa Z^\eta]\mathscr{E}^{(M)}(U, Z) = [U^{W-\kappa} Z^\eta]\mathscr{E}^{(M)}(U, Z)$ for all $0 \le \kappa \le W$ and all $\eta \ge 0$.

Proof. At $Z = 1$, $\operatorname{tr}(T_w^{(M)}(U, 1)^a) = (\operatorname{tr} T_w^{(M)}(U, 1))^a = (\sum_{r=0}^{M} U^{rw})^a$, and substituting this into (4.10) combines S_{comp} and S_{inc} into (5.1); setting $U = 1$ evaluates each factor at $M + 1$, yielding $(M+1)^{\sum_s |\mathcal{F}_s|} = (M+1)^{N_F}$.

For (b), on each orbit $O \in \mathcal{O}_s$ of length $a = a_O$, define the involution $\iota : \{0, \dots, M\}^a \to \{0, \dots, M\}^a$ by cyclic reversal and complementation, $(\iota(e))_i := M - e_{-i}$ ($i \in \mathbb{Z}/a\mathbb{Z}$). Then $\sum_{i=0}^{a-1}(M - (\iota(e))_i) = \sum_{i=0}^{a-1} e_{-i} = aM - \sum_{i=0}^{a-1}(M - e_i)$, while the hull statistic is preserved termwise up to reversal:

$$\sum_{i=0}^{a-1} \min\Big\{(\iota(e))_i,\, M - (\iota(e))_{i+1}\Big\} = \sum_{i=0}^{a-1} \min\{M - e_{-i},\, e_{-i-1}\} = \sum_{j=0}^{a-1} \min\{e_j,\, M - e_{j+1}\}$$

via $j = -i - 1 \in \mathbb{Z}/a\mathbb{Z}$. Thus $U^{awM} E_{a,w}^{(M)}(U^{-1}, Z) = E_{a,w}^{(M)}(U, Z)$. On each $f \in \mathcal{F}_s$ with $s \in S_{\text{inc}}$, substituting $r' = M - r$ gives $U^{Mm_s \deg f} \sum_{r=0}^{M} U^{-rm_s \deg f} = \sum_{r'=0}^{M} U^{r'm_s \deg f}$. Multiplying across all orbits and incompatible factors and noting $\sum_{s \in S_{\text{comp}}} \sum_{O \in \mathcal{O}_s} a_O w_O M + \sum_{s \in S_{\text{inc}}} \sum_{f \in \mathcal{F}_s} m_s(\deg f) M = W$ yields (5.2). □

Theorem 5.2 (Characterisation and p-power invariance of k-Galois LCD codes)**.** *Let $\lambda \in \mathcal{A}^*$ have partition $\{1,\dots,N\} = S_{\mathrm{comp}} \sqcup S_{\mathrm{inc}}$, and let $n = n'p^t$ with $\gcd(n',p) = 1$ and $M = p^t \geq 1$.*

(a) (Generator characterisation.) *An exponent selection $e = (e^{(s)})_{s=1}^{N} \in \mathscr{C}^{(M)}$ defines a k-Galois LCD code ($\mathrm{Hull}_{\mathcal{A},k}(C(e)) = \{0\}$) if and only if for every compatible component $s \in S_{\mathrm{comp}}$ and every $\tau_{s,k}$-orbit $O \in \mathcal{O}_s$, the restriction of $e^{(s)}$ to O is constant at an extreme value:*

$$\text{either } e_f = 0 \text{ for all } f \in O \quad \text{or} \quad e_f = M \text{ for all } f \in O,$$

equivalently $g_{e^{(s)}}(x) = g_{s,0}(x)^M$ where $\langle g_{s,0}(x)\rangle$ is a simple-root k-Galois LCD $\lambda_{s,0}$-constacyclic code of length n' over K_s, while $e^{(s)} \in \{0,\dots,M\}^{\mathcal{F}_s}$ is arbitrary for $s \in S_{\mathrm{inc}}$.

(b) (LCD generating polynomial and Jordan structure.) *The code-dimension enumerator of all k-Galois LCD λ-constacyclic codes is*

$$\mathscr{E}^{(M)}(U,0) = \left(\prod_{s\in S_{\mathrm{comp}}}\prod_{O\in\mathcal{O}_s}\left(1+U^{Ma_Ow_O}\right)\right)\left(\prod_{s\in S_{\mathrm{inc}}}\prod_{f\in\mathcal{F}_s}\frac{1-U^{(M+1)m_s\deg f}}{1-U^{m_s\deg f}}\right). \tag{5.3}$$

Moreover, the unweighted $(M+1)\times(M+1)$ LCD transition matrix $L_M := T_1^{(M)}(1,0)$ with entries $(L_M)_{\alpha,\beta} = \mathbf{1}_{\{\alpha=0 \text{ or } \beta=M\}}$ has rank 2 for every $M \geq 1$, characteristic polynomial $\det(tI_{M+1} - L_M) = t^{M-1}(t-1)^2$, Jordan block $\left(\begin{smallmatrix}1 & M\\ 0 & 1\end{smallmatrix}\right)$ on its 2-dimensional image, and $\mathrm{tr}(L_M^a) = 2$ for all $M \geq 1$ and $a \geq 1$.

(c) (Purely compatible p-power invariance.) *Whenever $S_{\mathrm{inc}} = \emptyset$ (in particular for all cyclic and negacyclic codes over $\mathcal{A}$), the total number of k-Galois LCD λ-constacyclic codes of length $n = n'p^t$ over $\mathcal{A}$ is*

$$\mathscr{H}^{(M)}(0) = \mathscr{E}^{(M)}(1,0) = 2^{\sum_{s=1}^{N}|\mathcal{O}_s|}, \tag{5.4}$$

completely independent of $t = v_p(n) \geq 0$, whereas the uniform probability that a random λ-constacyclic code is k-Galois LCD equals $2^{\sum_{s=1}^{N}|\mathcal{O}_s|}(p^t+1)^{-N_F}$.

Proof. By (4.4), $\eta_k(C(e)) = 0$ if and only if $\min\{e_{O,i}, M - e_{O,i+1}\} = 0$ for every $s \in S_{\mathrm{comp}}$, every $O \in \mathcal{O}_s$, and every $i \in \mathbb{Z}/a_O\mathbb{Z}$. On a fixed cycle of length $a = a_O$, the condition $\min\{e_i, M - e_{i+1}\} = 0$ reads: whenever $e_i \geq 1$, one must have $e_{i+1} = M$. Hence if $e_{i_0} \geq 1$ for at least one index $i_0 \in \mathbb{Z}/a\mathbb{Z}$, iterating a times forces $e_{i_0+1} = M \geq 1$, $e_{i_0+2} = M \geq 1$, and so on, giving $e_i = M$ for all $i \in \mathbb{Z}/a\mathbb{Z}$; otherwise $e_i = 0$ for all $i \in \mathbb{Z}/a\mathbb{Z}$. Conversely, both constant words 0^a and M^a give $\min\{e_i, M - e_{i+1}\} = 0$ everywhere and contribute U^{aw_OM} and 1 respectively to $\mathscr{E}^{(M)}(U,0)$, proving (a) and (5.3).

For the spectral statement in (b), write $v_1 := e_0 = (1,0,\dots,0)^{\mathsf{T}}$ and $v_2 := \sum_{\beta=1}^{M} e_\beta = (0,1,\dots,1)^{\mathsf{T}}$ in $\mathbb{Q}^{M+1}$. Columns $0,\dots,M-1$ of L_M all equal v_1 and column M equals $v_1 + v_2$, so $\mathrm{im}(L_M) = \mathrm{span}\{v_1, v_2\}$ has dimension 2, $\dim\ker(L_M) = M - 1$, $L_Mv_1 = v_1$, and

$L_M v_2 = (M-1)v_1 + (v_1+v_2) = Mv_1 + v_2$. Thus L_M acts as $\left(\begin{smallmatrix} 1 & M \\ 0 & 1 \end{smallmatrix}\right)$ on $\mathrm{span}\{v_1, v_2\}$ and nilpotently (0) on a supplement, yielding $\det(tI_{M+1} - L_M) = t^{M-1}(t-1)^2$ and $\mathrm{tr}(L_M^a) = 1^a + 1^a = 2$. Evaluating (5.3) at $U = 1$ with $S_{\mathrm{inc}} = \emptyset$ gives (5.4) and the stated density inside $|\mathscr{C}^{(M)}| = (p^t+1)^{N_F}$. □

Remark 5.3. At $N = 1$, $m_1 = 1$, $\lambda = 1$, and $k = 0$, Theorem 5.2(c) recovers the repeated-root Euclidean cyclic LCD count over $\mathbb{F}_q$ of Pang, Zhu, and Li [36] (and at $n = 2\ell^m p^t$, $k = 0$ aligns with Chen, Dinh, and Liu [35]), while extending it simultaneously to arbitrary k-Galois pairings, general constacyclic twists (showing that incompatibility switches the scaling from $2^{|\mathcal{O}_s|}$ to $(p^t+1)^{|\mathcal{F}_s|}$), non-split square-free affine algebras $\mathcal{A}$, and the full bivariate distribution around $\eta_k = 0$ illustrated in Figure 4.

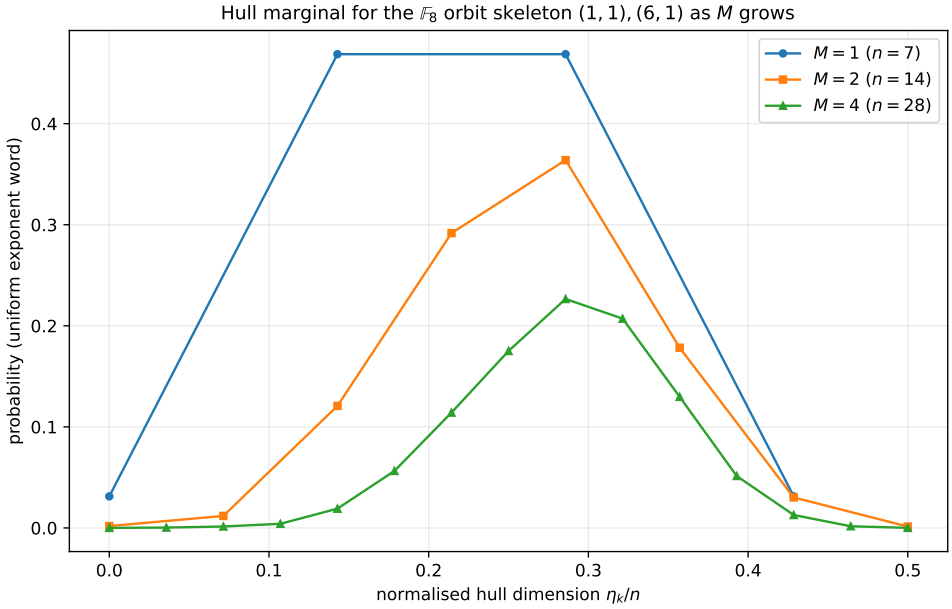


Figure 4: Evolution of the normalised marginal 1-Galois hull distribution $\mathbb{P}(\eta_1(C) = \eta)$ versus η/n along the 2-power multiplicity tower $M = 1, 2, 4$ ($n = 7, 14, 28$) for the cyclic $\mathbb{F}_8$ orbit skeleton $(1,1), (6,1)$: while the absolute LCD count at $\eta = 0$ stays locked at $2^2 = 4$ across all M, probability mass concentrates steadily around the asymptotic mean ratio $\eta/n \approx 19/63$ ($M = 2$) and $\to 13/42$ as $M \to \infty$.

5.3. Exact single-orbit support trapezoid and hull spectrum

Theorem 5.4 (Exact single-orbit support trapezoid)**.** *For every multiplicity $M \geq 1$, orbit length $a \geq 1$, and weight $w \geq 1$, the support of the single-orbit enumerator $E_{a,w}^{(M)}(U, Z) = \mathrm{tr}(T_w^{(M)}(U, Z)^a)$ is the exact lattice polygon*

$$\mathrm{Supp}\left(E_{a,w}^{(M)}\right) = \left\{(wR,\, wH) \in \mathbb{Z}_{\geq 0}^2 : 0 \leq R \leq aM,\ \ \min\{R,\, M,\, aM-R\} \leq H \leq \min\{R,\, aM-R\}\right\}. \tag{5.5}$$

In particular, the number of nonzero monomials in $E^{(M)}_{a,w}(U,Z)$ equals

$$\left|\mathrm{Supp}(E^{(M)}_{a,w})\right| = \begin{cases} M+1, & a=1, \\ aM+1+\left\lfloor \dfrac{(a-2)^2M^2}{4} \right\rfloor, & a\geq 2. \end{cases} \tag{5.6}$$

Proof. Scaling by w reduces to $w=1$. Put $x_i := M - e_i \in \{0,\dots,M\}$ ($i \in \mathbb{Z}/a\mathbb{Z}$), so $R(x) := \sum_{i=0}^{a-1} x_i \in \{0,\dots,aM\}$ and $H(x) := \sum_{i=0}^{a-1} \min\{M-x_i,\, x_{i+1}\}$. For $a=1$, $R = x_0$ and $H = \min\{M-R, R\} = \min\{R, M, M-R\}$, giving both (5.5) and cardinality $M+1$. For $a=2$, Corollary 4.7(b) gives $H(x_0,x_1) = \min\{R, 2M-R\} = \min\{R, M, 2M-R\}$ for every $0 \leq R \leq 2M$, yielding (5.5) and cardinality $2M+1 = 2M+1+\lfloor 0/4 \rfloor$.

Assume henceforth that $a \geq 3$. For the upper bound, summing the inequalities $\min\{M - x_i, x_{i+1}\} \leq x_{i+1}$ and $\min\{M-x_i, x_{i+1}\} \leq M-x_i$ around $\mathbb{Z}/a\mathbb{Z}$ gives $H(x) \leq \min\{R(x),\, aM - R(x)\}$.

For the lower bound $H(x) \geq \min\{R(x),\, M,\, aM - R(x)\}$, note first that the reversal-complement involution $\tilde{x}_i := M - x_{-i}$ satisfies $R(\tilde{x}) = aM - R(x)$, $H(\tilde{x}) = H(x)$, $\min_i \tilde{x}_i = M - \max_i x_i$, and $\max_i \tilde{x}_i = M - \min_i x_i$. Replacing x by $\tilde{x}$ if necessary, we may assume $x_{\min} + x_{\max} \leq M$, where $x_{\min} := \min_i x_i$ and $x_{\max} := \max_i x_i$. Write each edge minimum as the threshold sum (illustrated in Figure 5)

$$\min\{M - x_i,\, x_{i+1}\} = \sum_{r=1}^{M} \mathbf{1}_{\{x_i \leq M-r \text{ and } x_{i+1} \geq r\}}.$$

If $x_{\max} \leq \lfloor M/2 \rfloor$, then $x_i + x_{i+1} \leq 2x_{\max} \leq M$ for all i, so $\min\{M - x_i, x_{i+1}\} = x_{i+1}$ everywhere and $H(x) = R(x)$. Suppose instead that $x_{\max} \geq \lfloor M/2 \rfloor + 1$, set $K := M - x_{\max} \in \{0,\dots,\lfloor (M-1)/2 \rfloor\}$, and choose vertices $p, q \in \mathbb{Z}/a\mathbb{Z}$ with $x_p = x_{\min} \leq K$ and $x_q = x_{\max} = M - K$.

- For every $1 \leq r \leq K$, we have $x_i \leq x_{\max} = M - K \leq M - r$ for all i, so the r-th threshold sum around $\mathbb{Z}/a\mathbb{Z}$ equals $\#\{i : x_{i+1} \geq r\}$; summing over $1 \leq r \leq K$ contributes $\sum_{i=0}^{a-1} \min\{x_i, K\}$.

- Pair each remaining integer $K+1 \leq r \leq \lfloor M/2 \rfloor$ with $r' := M+1-r > r$, partitioning $\mathbb{Z}/a\mathbb{Z}$ into $L^{(r)} := \{i : x_i \leq r-1\}$, $C^{(r)} := \{i : r \leq x_i \leq M-r\}$, and $H^{(r)} := \{i : x_i \geq M+1-r\}$. Because $x_p \leq K \leq r-1$ and $x_q = M - K \geq M+1-r$, both $L^{(r)}$ and $H^{(r)}$ are non-empty. Subtracting the zero cycle sum of $\Phi^{(r)}(x_{i+1}) - \Phi^{(r)}(x_i)$ with $\Phi^{(r)} := \mathbf{1}_{H^{(r)}}$, the paired edge weight $c_r(i) := \mathbf{1}_{\{x_i \leq M-r,\, x_{i+1} \geq r\}} + \mathbf{1}_{\{x_i \leq r-1,\, x_{i+1} \geq M+1-r\}}$ satisfies $c_r(i) - (\Phi^{(r)}(x_{i+1}) - \Phi^{(r)}(x_i)) \geq 0$ on all nine state transitions, equals $+1$ on every exit from $H^{(r)}$ ($H^{(r)} \to L^{(r)} \cup C^{(r)}$), equals $+1$ on $L^{(r)} \to H^{(r)}$, and equals $+1$ on $L^{(r)} \to C^{(r)}$. Since $L^{(r)}$ and $H^{(r)}$ are both non-empty, traversing $\mathbb{Z}/a\mathbb{Z}$ leaves $H^{(r)}$ at least once ($+1$) and re-enters $H^{(r)}$ from $L^{(r)}$ either directly ($L^{(r)} \to H^{(r)}$, $+1$) or via $C^{(r)}$ preceded by $L^{(r)} \to C^{(r)}$ ($+1$), forcing $\sum_{i=0}^{a-1} c_r(i) \geq 2$. If M is odd, the unpaired middle threshold $r = (M+1)/2$ likewise has non-empty $L^{(r)}$ and $H^{(r)}$ ($C^{(r)} = \emptyset$), forcing at least one $L^{(r)} \to H^{(r)}$ edge ($+1$).

Combining all levels gives

$$H(x) \geq \sum_{i=0}^{a-1} \min\{x_i, K\} + 2\Big(\lfloor M/2 \rfloor - K\Big) + (M \bmod 2) = (M - 2K) + \sum_{i=0}^{a-1} \min\{x_i, K\}.$$

Separating vertex q (where $\min\{x_q, K\} = K$) rewrites the right-hand side as $(M-K) + \sum_{i \neq q} \min\{x_i, K\}$. If $x_i \leq K$ for all $i \neq q$ this equals $x_q + \sum_{i \neq q} x_i = R(x)$; if $x_{i_1} \geq K$ for at least one $i_1 \neq q$ it is at least $(M-K)+K = M$. In every case $H(x) \geq \min\{R(x), M, aM - R(x)\}$.

It remains to show that every integer $H \in [\min\{R, M, aM-R\}, \min\{R, aM-R\}]$ is attained for fixed $0 \leq R \leq aM$. By symmetry we may assume $R \leq \lfloor aM/2 \rfloor$. If $R \leq M$, the interval reduces to $\{R\}$, attained by $x = (R, 0, \ldots, 0)$. Suppose $M < R \leq \lfloor aM/2 \rfloor$. Write $R = q_0 M + r_0$ with $q_0 \geq 1$, $0 \leq r_0 < M$; since $R < aM$, the packed vector $x^{\min} := (M, \ldots, M, r_0, 0, \ldots, 0) \in \{0, \ldots, M\}^a$ has at least one zero coordinate and satisfies $H(x^{\min}) = M$ (only the closing $0 \to M$ edge contributes M). Define capacities $\mathrm{cap}_i := \lfloor M/2 \rfloor + \mathbf{1}_{\{M \text{ odd}, i \text{ even}, i < a-1\}}$ for $0 \leq i < a$. Readily $\sum_{i=0}^{a-1} \mathrm{cap}_i = \lfloor aM/2 \rfloor \geq R$ and $\mathrm{cap}_i + \mathrm{cap}_{i+1} \leq M$ for all $i \in \mathbb{Z}/a\mathbb{Z}$, so choosing any $x^{\max} \in \{0, \ldots, M\}^a$ with $0 \leq x_i^{\max} \leq \mathrm{cap}_i$ and $\sum_i x_i^{\max} = R$ gives $x_i^{\max} + x_{i+1}^{\max} \leq M$ everywhere and hence $H(x^{\max}) = R$.

Now transform $x^{\min}$ into $x^{\max}$ along the path 0–1–$\cdots$–$(a-1)$ by elementary adjacent unit transfers $(x_j, x_{j+1}) \mapsto (x_j \mp 1, x_{j+1} \pm 1)$ preserving $x \in \{0, \ldots, M\}^a$: matching coordinates from left to right, whenever $x_u > x_u^{\max}$ let $r > u$ be minimal with $x_r < M$ and slide one unit across $(r-1 \to r), \ldots, (u \to u+1)$, and symmetrically via the minimal $r > u$ with $x_r > 0$ when $x_u < x_u^{\max}$. Because $a \geq 3$, vertices $j-1, j, j+1$ are distinct at every single step $(x_j, x_{j+1}) \mapsto (x_j - 1, x_{j+1} + 1)$, and the three affected circular edges change by

$$\begin{aligned}
\Delta H = &\Big(\min\{M - x_{j-1}, x_j - 1\} - \min\{M - x_{j-1}, x_j\}\Big) \\
&+ \Big(\min\{M - x_j + 1, x_{j+1} + 1\} - \min\{M - x_j, x_{j+1}\}\Big) \\
&+ \Big(\min\{M - x_{j+1} - 1, x_{j+2}\} - \min\{M - x_{j+1}, x_{j+2}\}\Big) \in \{-1, 0\} + \{+1\} + \{-1, 0\} = \{-1, 0, +1\}.
\end{aligned}$$

Since H starts at M, ends at R, and changes by at most 1 at every step, every integer in $[M, R]$ is attained along the walk, proving (5.5).

Finally, summing slice heights $\min\{R, aM-R\} - \min\{R, M, aM-R\} + 1 = 1 + (\min\{R, aM-R\} - M)_+$ over $0 \leq R \leq aM$ gives $aM + 1 + \sum_{s=0}^{L} \min\{s, L-s\}$ with $L := (a-2)M$, and $\sum_{s=0}^{L} \min\{s, L-s\} = \lfloor L^2/4 \rfloor$ yields (5.6) (cf. Figure 6). □

Corollary 5.5 (Global attainable k-Galois hull spectrum)**.** *For any unit $\lambda \in \mathcal{A}^*$ and $0 \leq k < e$, the set of $\mathbb{F}_q$-dimensions attained by $\mathrm{Hull}_{\mathcal{A},k}(C)$ as C ranges over all λ-constacyclic codes of length $n = n'p^t$ ($M = p^t$) over $\mathcal{A}$ is the Minkowski sum*

$$\Big\{\eta_k(C) : C \in \mathscr{C}^{(M)}\Big\} = \sum_{s \in S_{\mathrm{comp}}} \sum_{O \in \mathcal{O}_s} w_O \cdot \Big\{0, 1, \ldots, \lfloor a_O M/2 \rfloor\Big\}, \tag{5.7}$$

understood as $\{0\}$ when $S_{\mathrm{comp}} = \emptyset$. In particular, $\max_C \eta_k(C) = \sum_{s \in S_{\mathrm{comp}}} \sum_{O \in \mathcal{O}_s} w_O \lfloor a_O M/2 \rfloor$.

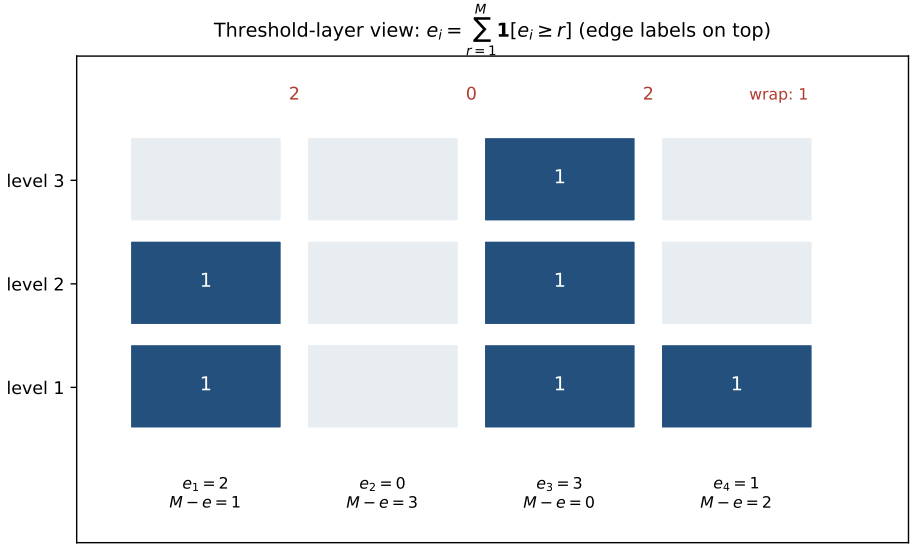


Figure 5: Binary threshold-layer decomposition $e_i = \sum_{r=1}^{M} \mathbf{1}_{\{e_i \geq r\}}$ underlying the lower-bound proof of Theorem 5.4, shown for $a = 4$, $M = 3$, and $e = (2, 0, 3, 1)$ with cyclic edge contributions $(2, 0, 2, 1)$.

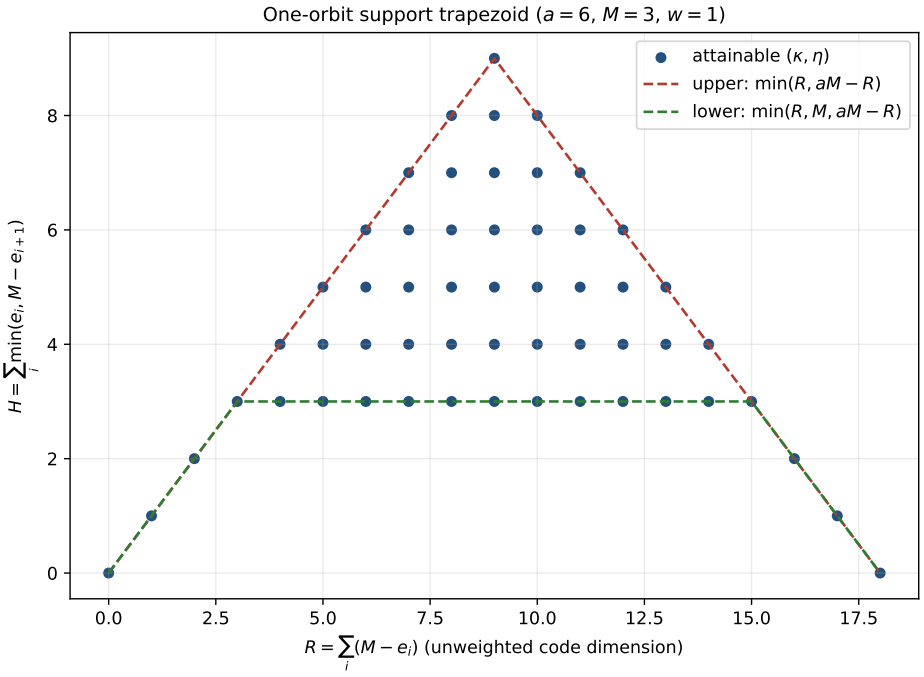


Figure 6: Exact single-orbit support polygon $\mathrm{Supp}(E_{6,1}^{(3)})$ for $a = 6$, $M = 3$, $w = 1$: every lattice point (R, H) between the lower floor $\min\{R, 3, 18 - R\}$ (green dashed) and upper tent $\min\{R, 18 - R\}$ (red dashed) is attained, giving $6 \cdot 3 + 1 + \lfloor 4^2 \cdot 3^2/4 \rfloor = 55$ distinct monomials.

Proof. Because every coefficient of $\mathrm{tr}(T_{w_O}^{(M)}(1,Z)^{a_O})$ is a non-negative integer, the support of the product (4.10) at $U=1$ is the Minkowski sum of the supports of its factors. On each single orbit (a_O, w_O), projecting (5.5) onto the second coordinate gives $\bigcup_{R=0}^{a_O M}[w_O \min\{R, M, a_O M - R\},\, w_O \min\{R, a_O M - R\}] = w_O\{0,1,\dots,\lfloor a_O M/2\rfloor\}$, because $R = 0,1,\dots,\min\{M,\lfloor a_O M/2\rfloor\}$ covers $\{0,\dots,\min\{M,\lfloor a_O M/2\rfloor\}\}$ while $R=\lfloor a_O M/2\rfloor$ covers $[\min\{M,\lfloor a_O M/2\rfloor\},\,\lfloor a_O M/2\rfloor]$. □

5.4. Exact moments, zero global covariance, and conditional laws

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Theorem 5.6 (Exact mean, variance, and covariance of (κ,η_k))**.** *Endow $\mathscr{C}^{(M)}$ with the uniform measure $\mathbb{P}(C(e)) = (M+1)^{-N_F}$, and write $M = p^t \geq 1$.*

(a) (Code dimension and global covariance.)

$$\mathbb{E}[\kappa] = \frac{W}{2}, \qquad \mathrm{Var}(\kappa) = \frac{M(M+2)}{12}\sum_{s=1}^{N}\sum_{f\in\mathcal{F}_s} m_s^2(\deg f)^2, \qquad \mathrm{Cov}(\kappa,\eta_k) = 0. \tag{5.8}$$

(b) (Hull mean and variance across all orbit lengths.)

$$\mathbb{E}[\eta_k] = \sum_{s\in S_{\mathrm{comp}}}\left(\sum_{\substack{O\in\mathcal{O}_s\\ a_O=1}} w_O\,\frac{\lfloor M^2/4\rfloor}{M+1} + \sum_{\substack{O\in\mathcal{O}_s\\ a_O\geq 2}} a_O w_O\,\frac{M(2M+1)}{6(M+1)}\right), \tag{5.9}$$

and

$$\mathrm{Var}(\eta_k) = \sum_{s\in S_{\mathrm{comp}}}\sum_{O\in\mathcal{O}_s} w_O^2\,\mathcal{V}_M(a_O), \tag{5.10}$$

where the unweighted orbit variance function $\mathcal{V}_M(a)$ is given cleanly across the three correlation regimes $a=1$, $a=2$, and $a\geq 3$ by

$$\mathcal{V}_M(a) = \begin{cases} \dfrac{M(M+2)(M^2+2M+4)}{48(M+1)^2}, & a=1,\ M \text{ even},\\[2mm] \dfrac{(M-1)(M+3)}{48}, & a=1,\ M \text{ odd},\\[2mm] \dfrac{M(M+2)(M^2+2M+3)}{18(M+1)^2}, & a=2,\\[2mm] \dfrac{a\,M(M+2)(M^2+2M+2)}{60(M+1)^2}, & a\geq 3. \end{cases} \tag{5.11}$$

At $M=1$ ($t=0$), (5.9)–(5.11) reduce to the simple-root variance constants $\mathcal{V}_1(1)=0$, $\mathcal{V}_1(2)=1/4$, and $\mathcal{V}_1(a)=a/16$ ($a\geq 3$). As $M=p^t\to\infty$ with fixed coprime part n', the normalised mean satisfies

$$\lim_{t\to\infty}\frac{\mathbb{E}[\eta_k]}{M} = \sum_{s\in S_{\mathrm{comp}}}\left(\frac{1}{4}\sum_{\substack{O\in\mathcal{O}_s\\ a_O=1}} w_O + \frac{1}{3}\sum_{\substack{O\in\mathcal{O}_s\\ a_O\geq 2}} a_O w_O\right).$$

Proof. Under the uniform measure on $\mathscr{C}^{(M)}$, the random exponents $(e_f)_{f\in\bigsqcup_s \mathcal{F}_s}$ are independent and uniformly distributed on $\{0,1,\dots,M\}$. Since $\mathbb{E}[M-e_f] = M/2$ and $\mathrm{Var}(M-e_f) = M(M+2)/12$, linearity and independence in (2.13) yield $\mathbb{E}[\kappa]$ and $\mathrm{Var}(\kappa)$. By Proposition 5.1(b), $(\kappa,\eta_k) \overset{d}{=} (W-\kappa,\eta_k)$, whence $\mathbb{E}[\kappa\eta_k] = \mathbb{E}[(W-\kappa)\eta_k] = W\mathbb{E}[\eta_k] - \mathbb{E}[\kappa\eta_k]$, so $\mathbb{E}[\kappa\eta_k] = \frac{W}{2}\mathbb{E}[\eta_k] = \mathbb{E}[\kappa]\mathbb{E}[\eta_k]$ and $\mathrm{Cov}(\kappa,\eta_k) = 0$.

Because distinct orbits are independent, both $\mathbb{E}[\eta_k]$ and $\mathrm{Var}(\eta_k)$ split additively over $O \in \mathcal{O}_s$. Let $X, Y, Z' \overset{\text{i.i.d.}}{\sim} \mathrm{Unif}\{0,\dots,M\}$ and put $g(u,v) := \min\{u, M-v\}$.

- *Regime $a = 1$:* $Y_1 := g(X,X) = \min\{X, M-X\}$ takes values $0,1,\dots,r-1$ twice each and r once when $M = 2r$ is even, and values $0,1,\dots,r$ twice each when $M = 2r+1$ is odd. Direct summation of $\sum x$ and $\sum x^2$ gives $\mathbb{E}[Y_1] = \lfloor M^2/4 \rfloor/(M+1)$ and the two parity formulas for $\mathcal{V}_M(1) = \mathrm{Var}(Y_1)$ in (5.11).

- *Single-edge moments ($a \geq 2$):* For $1 \leq r \leq M$, $\#\{(u,v) \in \{0,\dots,M\}^2 : g(u,v) \geq r\} = (M+1-r)^2$. Summing 1 and $2r-1$ against $(M+1-r)^2$ over $1 \leq r \leq M$ gives

  $$\mathbb{E}[g(X,Y)] = \frac{M(2M+1)}{6(M+1)}, \qquad \mathbb{E}[g(X,Y)^2] = \frac{M(M^2+M+1)}{6(M+1)}, \qquad \mathrm{Var}\Big(g(X,Y)\Big) = \frac{M(M+2)(2M}{36(M}$$

  Summing $\mathbb{E}[g(X_i, X_{i+1})]$ around $i \in \mathbb{Z}/a\mathbb{Z}$ gives (5.9).

- *Regime $a = 2$:* Direct summation over $0 \leq x, y \leq M$ yields $\sum_{x,y=0}^{M} g(x,y)g(y,x) = M^2(M-1)(M+1)/12$, so $\mathrm{Cov}(g(X,Y), g(Y,X)) = -M^2(M+2)^2/\Big(36(M+1)^2\Big)$ and

  $$\mathcal{V}_M(2) = \mathrm{Var}\Big(g(X,Y)+g(Y,X)\Big) = 2\Big(\mathrm{Var}(g(X,Y))+\mathrm{Cov}\Big(g(X,Y), g(Y,X)\Big)\Big) = \frac{M(M+2)(M^2+2M}{18(M+1)^2}$$

- *Regime $a \geq 3$:* Along the cycle $(X_0,\dots,X_{a-1})$ with $a \geq 3$, edges $g(X_i, X_{i+1})$ and $g(X_j, X_{j+1})$ share no variables (hence have zero covariance) whenever $j \notin \{i-1, i, i+1\}$ modulo a, while each of the a adjacent pairs $(g(X,Y), g(Y,Z'))$ satisfies, upon conditioning on $c := M - Y \in \{0,\dots,M\}$,

  $$\sum_{x=0}^{M} g(x,Y) = \frac{c(2M+1-c)}{2}, \qquad \sum_{z=0}^{M} g(Y,z) = \frac{(M-c)(M+1+c)}{2},$$

  whence $\mathbb{E}[g(X,Y)g(Y,Z')] = M(M-1)(11M^2+15M+6)/\Big(120(M+1)^2\Big)$ (vanishing at $M=1$ because $g(X,Y)=1$ forces $Y=0$ whereas $g(Y,Z')=1$ forces $Y=1$) and

  $$\mathrm{Cov}\Big(g(X,Y), g(Y,Z')\Big) = -\frac{M(M+2)(7M^2+14M+9)}{360(M+1)^2}.$$

  Note that adjacent edges are strictly negatively correlated for every $M \geq 1$. Therefore

  $$\mathcal{V}_M(a) = a\Big(\mathrm{Var}(g(X,Y)) + 2\,\mathrm{Cov}\Big(g(X,Y), g(Y,Z')\Big)\Big) = \frac{a\,M(M+2)(M^2+2M+2)}{60(M+1)^2}.$$

□

Table 3: Exact unweighted local expectation, covariance, and variance constants across prime-power multiplicities $M = p^t$ (Theorem 5.6).

M	$\mathbb{E}[g]$ $(a \geq 2)$	$\mathbb{E}[Y_1]$ $(a = 1)$	$\mathrm{Var}(g)$	$\mathrm{Cov}_{\mathrm{adj}}$	$\mathcal{V}_M(a)/a$ $(a \geq 3)$	$\mathcal{V}_M(2)$	$\mathcal{V}_M(1)$
1	1/4	0	3/16	−1/16	1/16	1/4	0
2	5/9	1/3	38/81	−13/81	4/27	44/81	2/9
3	7/8	1/2	55/64	−19/64	17/64	15/16	1/4
4	6/5	4/5	34/25	−59/125	52/125	36/25	14/25
5	55/36	1	2555/1296	−889/1296	259/432	665/324	2/3
8	68/27	16/9	3260/729	−1138/729	328/243	3320/729	140/81
9	57/20	2	2211/400	−3861/2000	3333/2000	561/100	2
16	88/17	64/17	4632/289	−1620/289	1392/289	4656/289	1752/289
25	425/52	6	101475/2704	−35505/2704	30465/2704	25425/676	14
27	495/56	13/2	136503/3136	−47763/3136	40977/3136	34191/784	65/4
32	1040/99	256/33	592688/9801	−207400/9801	59296/3267	593504/9801	24752/1089

Table 3 records exact rational values of the local moment constants across representative prime powers $M = p^t$.

Although $\mathrm{Cov}(\kappa, \eta_k) = 0$ globally, η_k depends strongly and concavely on the code dimension κ, as captured exact slice by slice via factorial-moment extraction (Figure 7).

Proposition 5.7 (Exact conditional moments given code dimension)**.** *For $j \in \{0, 1, 2\}$ and $0 \leq \kappa \leq W$, put $\Phi_j(\kappa) := [U^\kappa]\, \partial_Z^j \mathscr{E}^{(M)}(U, Z)\big|_{Z=1}$. Whenever $\Phi_0(\kappa) > 0$,*

$$\mathbb{E}[\eta_k \mid \kappa(C) = \kappa] = \frac{\Phi_1(\kappa)}{\Phi_0(\kappa)}, \qquad \mathrm{Var}(\eta_k \mid \kappa(C) = \kappa) = \frac{\Phi_2(\kappa) + \Phi_1(\kappa)}{\Phi_0(\kappa)} - \left(\frac{\Phi_1(\kappa)}{\Phi_0(\kappa)}\right)^2, \quad (5.12)$$

and both functions satisfy the reflection identity $F(W - \kappa) = F(\kappa)$ for all attainable κ. Moreover, Φ_0, Φ_1, Φ_2 are determined without forming full bivariate polynomials by evaluating (4.10) in the truncated jet ring $\mathbb{Z}[U][Z]/\langle (Z-1)^3 \rangle$.

Proof. Differentiating $\mathscr{E}^{(M)}(U, Z) = \sum_{\kappa,\eta} N_{\kappa,\eta} U^\kappa Z^\eta$ j times with respect to Z at $Z = 1$ gives $\Phi_0(\kappa) = \sum_\eta N_{\kappa,\eta}$, $\Phi_1(\kappa) = \sum_\eta \eta N_{\kappa,\eta}$, and $\Phi_2(\kappa) = \sum_\eta \eta(\eta-1) N_{\kappa,\eta}$, from which (5.12) follows via $\eta^2 = \eta(\eta-1) + \eta$. Symmetry in $\kappa \leftrightarrow W - \kappa$ follows by differentiating (5.2) at $Z = 1$, and expanding $Z = 1 + \varepsilon \pmod{\varepsilon^3}$ yields $\mathscr{E}^{(M)}(U, 1+\varepsilon) \equiv \sum_\kappa \Big(\Phi_0(\kappa) + \Phi_1(\kappa)\varepsilon + \frac{1}{2}\Phi_2(\kappa)\varepsilon^2\Big) U^\kappa$. □

5.5. Self-orthogonal, dual-containing, and self-dual code counts

Theorem 5.8 (Enumeration of extremal k-Galois duality classes)**.** *For $M \geq 1$, define $(M+1) \times (M+1)$ $\{0,1\}$-matrices on $\{0, \ldots, M\}$ by*

$$\left(A_{\mathrm{SO}}^{(M)}\right)_{\alpha,\beta} := \mathbf{1}_{\{\alpha+\beta \geq M\}}, \qquad \left(A_{\mathrm{DC}}^{(M)}\right)_{\alpha,\beta} := \mathbf{1}_{\{\alpha+\beta \leq M\}}, \qquad \left(A_{\mathrm{SD}}^{(M)}\right)_{\alpha,\beta} := \mathbf{1}_{\{\alpha+\beta = M\}}.$$

Let $\lambda \in \mathcal{A}^*$ have partition $\{1, \ldots, N\} = S_{\mathrm{comp}} \sqcup S_{\mathrm{inc}}$.

(a) *The numbers $N_{\mathrm{SO}}^{(M)}$ of k-Galois self-orthogonal codes ($C \subseteq D_{\mathcal{A},k}(C)$, equivalently $\eta_k(C) = \kappa(C)$) and $N_{\mathrm{DC}}^{(M)}$ of k-Galois dual-containing codes ($D_{\mathcal{A},k}(C) \subseteq C$, equivalently $\eta_k(C) = W - \kappa(C)$) satisfy*

$$N_{\mathrm{SO}}^{(M)} = N_{\mathrm{DC}}^{(M)} = \prod_{s \in S_{\mathrm{comp}}} \prod_{O \in \mathcal{O}_s} \mathrm{tr}\Big(\Big(A_{\mathrm{SO}}^{(M)}\Big)^{a_O}\Big), \quad (5.13)$$

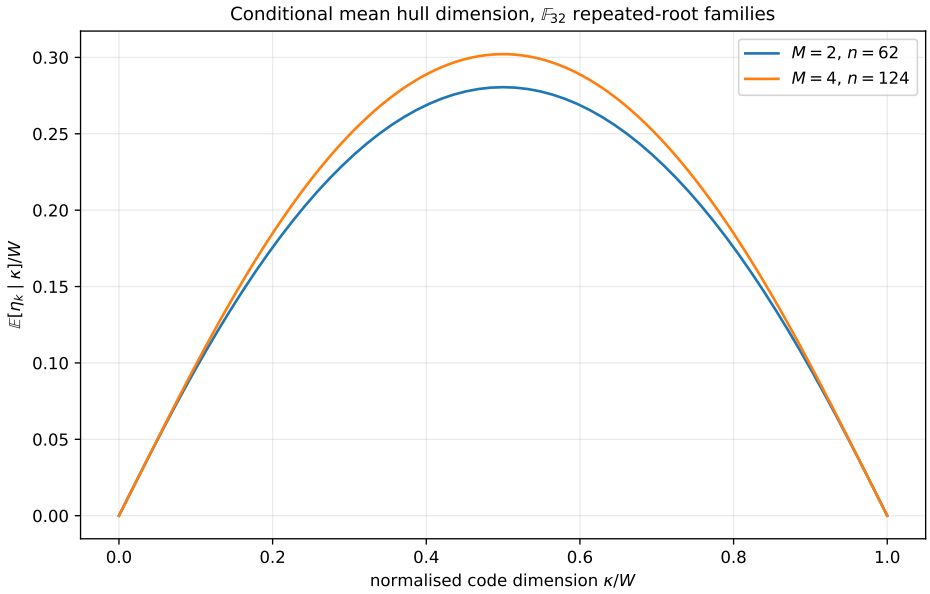


Figure 7: Exact conditional expectation $\mathbb{E}[\eta_1 \mid \kappa]/W$ versus normalised rate κ/W for cyclic codes over $\mathbb{F}_{32}$ at repeated-root lengths $n = 62$ ($M = 2$, $3^{31} \approx 6.18 \times 10^{14}$ codes) and $n = 124$ ($M = 4$, $5^{31} \approx 4.66 \times 10^{21}$ codes), symmetric around $\kappa/W = 1/2$ with global linear covariance $\mathrm{Cov}(\kappa, \eta_1) = 0$.

and their $\mathbb{F}_q$-dimension enumerators equal $[t^0]\,\mathscr{E}^{(M)}(Ut, t^{-1})$ and $[t^W]\,\mathscr{E}^{(M)}(Ut, t)$ respectively. For $a_O = 1$ and $a_O = 2$, $\mathrm{tr}(A_{\mathrm{SO}}^{(M)}) = \lceil (M+1)/2 \rceil$ and $\mathrm{tr}\big((A_{\mathrm{SO}}^{(M)})^2\big) = \binom{M+2}{2}$, while at $M = 1$ the traces $\mathrm{tr}\big((A_{\mathrm{SO}}^{(1)})^a\big) = L_a$ ($L_1 = 1$, $L_2 = 3$, $L_a = L_{a-1} + L_{a-2}$) reproduce the classical Lucas numbers.

(b) *The number $N_{\mathrm{SD}}^{(M)}$ of k-Galois self-dual λ-constacyclic codes ($C = D_{\mathcal{A},k}(C)$, equivalently $\kappa(C) = \eta_k(C) = W/2$) is given in closed form by*

$$N_{\mathrm{SD}}^{(M)} = \begin{cases} 0, & \text{if } S_{\mathrm{inc}} \neq \emptyset \text{ or } \big(M \text{ is odd and } \exists\, s \in S_{\mathrm{comp}},\, O \in \mathcal{O}_s \text{ wi} \\ (M+1)^{\#\{(s,O)\,:\, s \in S_{\mathrm{comp}},\, O \in \mathcal{O}_s,\, a_O \text{ even}\}}, & \text{otherwise.} \end{cases} \tag{5.14}$$

In particular, passing to repeated roots $t \geq 1$ in characteristic $p = 2$ ($M = 2^t$ even) unlocks self-dual constacyclic codes whenever self-reciprocal or odd-length orbits exist on compatible components, whereas in odd characteristic $p \geq 3$ ($M = p^t$ odd) odd-length orbits remain a permanent obstruction for every $t \geq 0$.

Proof. Since $\mathrm{Hull}_{\mathcal{A},k}(C) = C \cap D_{\mathcal{A},k}(C)$ and $\dim_{\mathbb{F}_q} D_{\mathcal{A},k}(C) = W - \kappa(C)$, we have $C \subseteq D_{\mathcal{A},k}(C) \iff \eta_k(C) = \kappa(C)$, $D_{\mathcal{A},k}(C) \subseteq C \iff \eta_k(C) = W - \kappa(C)$, and $C = D_{\mathcal{A},k}(C) \iff \eta_k(C) = \kappa(C) = W/2$, yielding the generating-function extractions from $\mathscr{E}^{(M)}(U, Z)$. Directly at the generator level, Proposition 3.8(b) pins every $s \in S_{\mathrm{inc}}$ to the unique choice $e_f = M$ for self-orthogonality (1 factor), $e_f = 0$ for dual containment (1 factor), and 0 choices for self-duality. On each $s \in S_{\mathrm{comp}}$, comparing exponents in (2.8) and (3.12) shows that $C_s(e) \subseteq D_k(C_s(e))$ holds if and only if $M - e_{O,i-1} \leq e_{O,i}$, i.e. $e_{O,i-1} + e_{O,i} \geq M$, along every orbit $O \in \mathcal{O}_s$, counted by $\mathrm{tr}\big((A_{\mathrm{SO}}^{(M)})^{a_O}\big)$. Conjugating $A_{\mathrm{SO}}^{(M)}$ by the permutation involution

$J := A_{\mathrm{SD}}^{(M)}$ ($J_{\alpha,\beta} = \delta_{\alpha,M-\beta}$) gives $JA_{\mathrm{SO}}^{(M)}J = A_{\mathrm{DC}}^{(M)}$, proving $\mathrm{tr}\big((A_{\mathrm{SO}}^{(M)})^{a_O}\big) = \mathrm{tr}\big((A_{\mathrm{DC}}^{(M)})^{a_O}\big)$ and (5.13). For $a = 1$, $\#\{\alpha \in \{0,\dots,M\} : 2\alpha \geq M\} = \lceil (M+1)/2 \rceil$; for $a = 2$, $\#\{(\alpha,\beta) : \alpha+\beta \geq M\} = \sum_{r=0}^{M}(r+1) = \binom{M+2}{2}$.

Finally, self-duality on orbit O holds if and only if $e_{O,i-1} + e_{O,i} = M$ for all $i \in \mathbb{Z}/a_O\mathbb{Z}$, counted by $\mathrm{tr}(J^{a_O})$. Since $J^2 = I_{M+1}$, $\mathrm{tr}(J^{a_O}) = M+1$ when a_O is even, while when a_O is odd $\mathrm{tr}(J^{a_O}) = \mathrm{tr}(J) = \#\{\alpha : 2\alpha = M\} = \mathbf{1}_{\{M \text{ even}\}}$, giving (5.14). □

Table 4 lists exact single-orbit counts across $1 \leq M \leq 5$ and representative orbit lengths a.

Table 4: Exact single-orbit counts of total codes $(M+1)^a$, LCD codes, self-orthogonal / dual-containing codes ($N_{\mathrm{SO}} = N_{\mathrm{DC}}$), and self-dual codes across $1 \leq M \leq 5$ (Theorems 5.2 and 5.8).

a	M	$(M+1)^a$	LCD	SO / DC	SD	a	M	$(M+1)^a$	LCD	SO / DC	SD
1	1	2	2	1	0	4	1	16	2	7	2
1	2	3	2	2	1	4	2	81	2	26	3
1	3	4	2	2	0	4	3	256	2	70	4
1	4	5	2	3	1	4	4	625	2	155	5
1	5	6	2	3	0	4	5	1296	2	301	6
2	1	4	2	3	2	5	1	32	2	11	0
2	2	9	2	6	3	5	2	243	2	57	1
2	3	16	2	10	4	5	3	1024	2	197	0
2	4	25	2	15	5	5	4	3125	2	533	1
2	5	36	2	21	6	5	5	7776	2	1223	0
3	1	8	2	4	0	6	1	64	2	18	2
3	2	27	2	11	1	6	2	729	2	129	3
3	3	64	2	23	0	6	3	4096	2	571	4
3	4	125	2	42	1	6	4	15625	2	1884	5
3	5	216	2	69	0	10	5	60466176	2	1508401	6

6. Computation and examples

We next translate Theorem 4.6 and Proposition 5.7 into exact polynomial-time dynamic-programming algorithms (§6.1), document our independent linear-algebraic and quotient-ring verification suite (§6.2), and analyse benchmark code families spanning characteristics $p \in \{2,3,5\}$, non-cyclic twists, non-split affine algebras, and state spaces exceeding 4.6×10^{21} codes (§6.3).

6.1. Dynamic-programming evaluation and arithmetic complexity

Brute-force enumeration over $\mathscr{C}^{(M)}$ inspects $(M+1)^{N_F}$ codes, which is exponential in both N_F and $t = v_p(n)$. By contrast, (4.10) computes $\mathscr{E}^{(M)}(U,Z)$ without listing individual codes.

Proposition 6.1 (Polynomial-time dynamic programming for $\mathscr{E}^{(M)}(U,Z)$)**.** *Given the orbit multisets* $\{(a_O, w_O) : O \in \mathcal{O}_s\}$ *($s \in S_{\mathrm{comp}}$) and factor degrees* $\{\deg f : f \in \mathcal{F}_s\}$ *($s \in S_{\mathrm{inc}}$), with ambient dimension* $W = n\sum_{s=1}^{N} m_s$ *and multiplicity* $M = p^t$*:*

(a) (Full bivariate table.) *Initialising $G(U,Z) := 1$ and, for each compatible orbit (a_O, w_O), summing over initial states $s_0 \in \{0,\dots,M\}$ the a_O-step DP array $S_r[u]$ ($0 \le r \le a_O$, $0 \le u \le M$) updated by*

$$S_{r+1}[\beta] \;+=\; U^{w_O(M-\beta)}\, Z^{w_O \min\{\alpha,\, M-\beta\}}\, S_r[\alpha] \qquad (0 \le \alpha, \beta \le M),$$

followed by f-wise multiplication across $s \in S_{\mathrm{inc}}$, computes all nonzero coefficients of $\mathscr{E}^{(M)}(U,Z)$ in

$$O\Big(M^3 \sum_{s \in S_{\mathrm{comp}}} a_O \cdot |\mathcal{S}_{\max}|\Big) = O\big(M^3 N_F W^2\big) \tag{6.1}$$

operations on exact $O(N_F \log M)$-bit integers, where $|\mathcal{S}_{\max}| \le (W+1)(\lfloor W/2 \rfloor + 1) = O(W^2)$ bounds the intermediate support size. Alternatively, binary exponentiation of $T_{w_O}^{(M)}(U,Z)^{a_O}$ evaluates each single-orbit trace in $O(\log a_O)$ polynomial-matrix multiplications on $(M+1)\times(M+1)$ matrices.

(b) (Truncated 3-jet for conditional moments.) *Carrying coefficients in $\mathbb{Z}[U][Z]/\langle (Z-1)^3 \rangle$ replaces bivariate tables of size $O(W^2)$ by triples of degree-$\le W$ univariate polynomials, computing $\Phi_0(\kappa), \Phi_1(\kappa), \Phi_2(\kappa)$ for all $0 \le \kappa \le W$ in $O(M^3 N_F W)$ integer operations.*

Proof. Fix an orbit (a, w). For each pinned start state $s_0 \in \{0,\dots,M\}$, induction on $r = 0,\dots,a$ shows that $S_r[u]$ equals $\sum U^{w\sum_{i=1}^{r}(M-e_i)} Z^{w\sum_{i=0}^{r-1}\min\{e_i, M-e_{i+1}\}} G(U,Z)$ summed over walks $s_0 = e_0 \to e_1 \to \cdots \to e_r = u$. Extracting $S_a[s_0]$ and summing over all $M+1$ choices of s_0 multiplies $G(U,Z)$ by $\mathrm{tr}(T_w^{(M)}(U,Z)^a)$ by Lemma 4.5. Each of the $\sum_O a_O \le N_F$ transitions executes $(M+1)$ start states $\times\, (M+1)^2$ pairs (α,β) across at most $|\mathcal{S}_{\max}|$ nonzero (κ,η) entries, giving (6.1). In $\mathbb{Z}[U][Z]/\langle (Z-1)^3\rangle$, only three Z-jet coordinates exist per $\kappa \in \{0,\dots,W\}$, reducing $|\mathcal{S}_{\max}|$ from $O(W^2)$ to $3(W+1) = O(W)$. □

6.2. Independent linear-algebraic and quotient-ring verification

To guard against any hidden algebra, indexing, or parity error, every formula in Sections 3–5 was cross-checked by an independent standard-library Python verification suite (`verify_rr.py`) requiring zero input from transfer matrices or orbit combinatorics:

1. *Arbitrary-degree Cantor–Zassenhaus factorisation and Gram-rank verification.* Over explicit prime-extension fields $K_s = \mathbb{F}_{p^d}$ across $p \in \{2,3,5\}$, $x^{n'} - \lambda_{s,0}$ is factored from scratch by distinct-degree and Cantor–Zassenhaus splitting (covering linear, quadratic, cubic, and quartic irreducibles). For every exponent tuple e, the monic divisor $g_e(x)$ is expanded directly in $K_s[x]$, a $(\dim_{K_s} C) \times n$ shift generator matrix B is formed in K_s^n, and $\dim_{K_s} \mathrm{Hull}_k(C_s(e))$ is computed directly from $\dim_{K_s} C - \mathrm{rank}_{K_s}(B\,\sigma_{s,k}(B)^{\mathsf{T}})$ via (3.14). Exhaustive runs verified complete joint histograms across eleven families (including all 2,187 codes over $\mathbb{F}_8$ at $n = 14$, degree-3 cubics over $\mathbb{F}_4$ at $n = 14$, and degree-4 quartics over $\mathbb{F}_3$ at $n = 24$), supplemented by seeded random subsamples at $n = 24, 30, 62$.
2. *Direct verification in raw quotient-polynomial coordinates.* For the non-split algebra $\mathcal{A} = \mathbb{F}_4[X]/\langle X(X^2+X+\omega)\rangle \cong \mathbb{F}_4 \times \mathbb{F}_{16}$ at repeated-root length $n = 10 = 5\cdot 2$ ($M = 2$, $k = 1$), codes were spanned directly inside $\mathcal{A}^{10} \cong \mathbb{F}_4^{30}$ in the monomial basis $\{1, X, X^2\}$ without passing to CRT coordinates, using ring multiplication modulo $X^3 + X^2 + \omega X$

and the intrinsic Frobenius $Z \mapsto Z^2$ on $\mathcal{A}$. Both the non-identity compatible unit $\lambda_0 \leftrightarrow (\omega, 1)$ ($\lambda \leftrightarrow (\omega^2, 1)$) and the mixed compatible/incompatible unit $\lambda_0 \leftrightarrow (\omega, \beta)$ ($\lambda \leftrightarrow (\omega^2, \beta^2)$, with $\beta^4 + \beta + 1 = 0$ in $\mathbb{F}_{16}$) matched Theorem 4.2 and Proposition 3.8 across all tested codes without discrepancy.

6.3. Worked benchmark families

Table 5 collects exact invariants across eleven representative repeated-root families; in accordance with Remark 2.3, every family records both the base root $\lambda_{s,0}$ and the length-n constacyclic shift $\lambda_s = \lambda_{s,0}^M$.

Table 5: Exact invariants across eleven repeated-root benchmark families ($n = n'M$, $M = p^t$). In all rows both $\lambda_{s,0}$ and $\lambda_s = \lambda_{s,0}^M$ are stated explicitly.

Algebra $\mathcal{A}$	(n', M, n)	k	λ_0	$\lambda = \lambda_0^M$	Orbits (a_O, w_O)	$\lvert\mathscr{C}^{(M)}\rvert$	$\lvert\mathrm{Supp}\rvert$	LCD	SO = DC	SD	$\mathbb{E}[\eta_k]$	$\mathrm{Var}(\eta_k)$
$\mathbb{F}_4$	$(5, 2, 10)$	1	1	1	$(1,1),(2,2)$	27	15	4	12	3	23/9	194/81
$\mathbb{F}_4$	$(5, 2, 10)$	1	ω	ω^2	$(1,1),(2,2)$	27	15	4	12	3	23/9	194/81
$\mathbb{F}_8$	$(7, 2, 14)$	1	1	1	$(1,1),(6,1)$	2,187	44	4	258	3	11/3	10/9
$\mathbb{F}_3$	$(4, 3, 12)$	0	1	1	$(1,1)^2,(1,2)$	64	33	8	8	0	2	3/2
$\mathbb{F}_9$	$(4, 3, 12)$	1	i	$-i$	$(2,2)$	16	7	2	10	4	7/2	15/4
$\mathbb{F}_9$	$(4, 3, 12)$	1	-1	-1	$(2,1)^2$	256	22	4	100	16	7/2	15/8
$\mathbb{F}_9$	$(8, 3, 24)$	1	1	1	$(1,1)^4,(2,1)^2$	65,536	142	64	1,600	0	11/2	23/8
$\mathbb{F}_{25}$	$(6, 5, 30)$	1	1	1	$(1,1)^6$	46,656	191	64	729	0	6	4
$\mathbb{F}_4 \times \mathbb{F}_{16}$	$(5, 2, 10)$	1	$(1,1)$	$(1,1)$	$(1,1),(2,2),(1,2),(4,2)$	6,561	123	16	624	9	23/3	458/81
$\mathbb{F}_{32}$	$(31, 2, 62)$	1	1	1	$(1,1),(10,1)^3$	3^{31}	916	16	70,639,674,082	27	17	14/3
$\mathbb{F}_{32}$	$(31, 4, 124)$	1	1	1	$(1,1),(10,1)^3$	5^{31}	3,531	16	$\approx 7.06 \times 10^{16}$	125	184/5	326/25

Example 6.2 (Long orbit and interior trapezoid states over $\mathbb{F}_8$)**.** Take $\mathcal{A} = \mathbb{F}_8$ ($p = 2$, $e = 3$, $m_1 = 1$), $k = 1$, $n = 14 = 7 \cdot 2^1$ ($n' = 7$, $M = 2$), and cyclic twist $\lambda_{1,0} = 1$, $\lambda_1 = \lambda_{1,0}^2 = 1$. Over $\mathbb{F}_8$, $x^7 - 1$ splits completely into seven linear factors, partitioned by $\tau_{1,1}$ into $(f_0) = (x + 1)$ of length $a_1 = 1$ ($w_1 = 1$) and one length-6 cycle O_2 ($a_2 = 6$, $w_2 = 1$). Theorem 4.6(b) yields all $3^7 = 2{,}187$ codes across 44 (κ, η_1) cells (Figure 8). In contrast to Example 4.8 ($a_O \le 2$), O_2 has $a_2 = 6 \ge 3$ and therefore populates all $\lfloor (6-2)^2 \cdot 2^2/4 \rfloor = 16$ strictly interior lattice points of $\mathrm{Supp}(E_{6,1}^{(2)})$ above the flat floor $H = 2$ (Theorem 5.4). Theorems 5.2, 5.6, and 5.8 give $N_{\mathrm{LCD}} = 2^2 = 4$ (identical to simple-root $n' = 7$), $\mathbb{E}[\eta_1] = 1/3 + 6(5/9) = 11/3$, $\mathrm{Var}(\eta_1) = 2/9 + 6(4/27) = 10/9$, $N_{\mathrm{SO}} = N_{\mathrm{DC}} = 2 \cdot 129 = 258$, and $N_{\mathrm{SD}} = 1 \cdot 3 = 3$ (at $(\kappa, \eta_1) = (7, 7)$, whereas $n' = 7$ has zero self-dual codes).

Example 6.3 (Odd-characteristic constacyclic versus negacyclic codes over $\mathbb{F}_9$)**.** Let $\mathcal{A} = \mathbb{F}_9 = \mathbb{F}_3(i)$ with $i^2 + 1 = 0$ ($p = 3$, $e = 2$, $m_1 = 1$), $k = 1$, and $n = 12 = 4 \cdot 3^1$ ($n' = 4$, $M = 3$). Compare two distinct compatible units:

(a) Take $\lambda_{1,0} = i$, so the length-12 constacyclic shift is $\lambda_1 = i^3 = -i \neq \lambda_{1,0}$. Since $i^{1+3^{2-1}} = i^4 = 1$, compatibility holds; $x^4 - i = (x^2 + x - i)(x^2 - x - i)$ factors into a single $\tau_{1,1}$-pair of degree 2, giving one orbit $(a_1, w_1) = (2, 2)$, $4^2 = 16$ codes across 7 cells, 2 LCD codes, 10 SO/DC codes, 4 self-dual codes, $\mathbb{E}[\eta_1] = 2 \cdot 2 \cdot (7/8) = 7/2$, and $\mathrm{Var}(\eta_1) = 2^2 \cdot (15/16) = 15/4$.

(b) Take $\lambda_{1,0} = -1$, so $\lambda_1 = (-1)^3 = -1$ (negacyclic). Now $x^4 + 1$ splits completely over $\mathbb{F}_9$ into four linear factors forming two $\tau_{1,1}$-pairs $(2,1),(2,1)$: there are $4^4 = 256$ codes

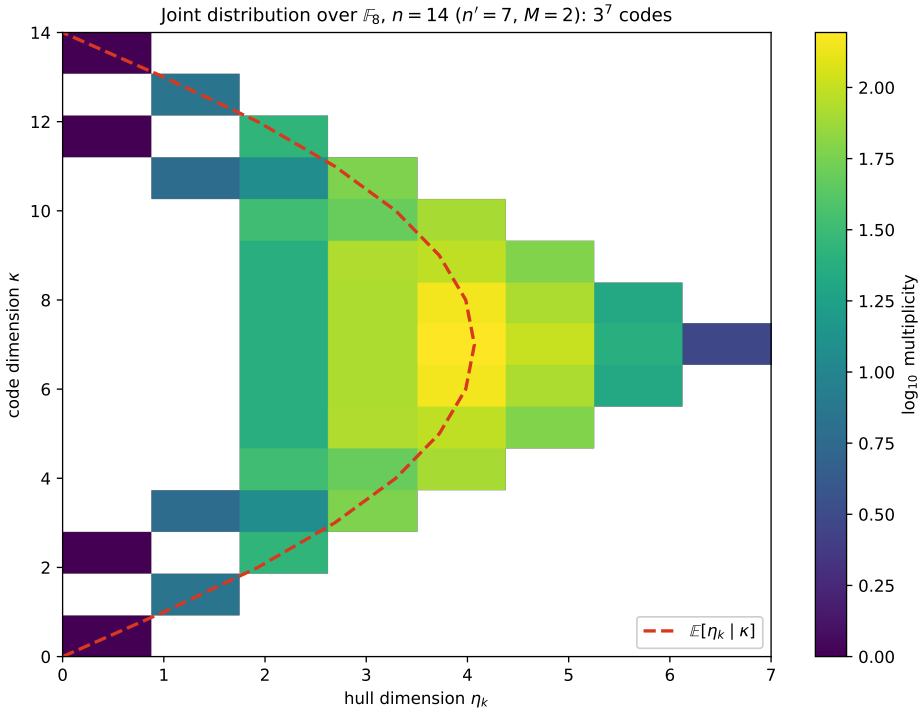


Figure 8: Exact joint code-dimension and 1-Galois-hull multiplicity map for all $3^7 = 2{,}187$ cyclic codes of repeated-root length $n = 14$ ($n' = 7$, $M = 2$) over $\mathbb{F}_8$, with the exact rational conditional expectation curve $\mathbb{E}[\eta_1 \mid \kappa]$ overlaid (peaking at $533/131 \approx 4.069$ at $\kappa = 7$).

across 22 cells, 4 LCD codes, 100 SO/DC codes, and $4^2 = 16$ self-dual codes. Notice that $\mathbb{E}[\eta_1] = 2 \cdot (7/4) = 7/2$ coincides with (a), whereas $\mathrm{Var}(\eta_1) = 2 \cdot (15/16) = 15/8$ is halved because w_O^2 drops from 4 to $1 + 1 = 2$. By contrast, at cyclic shift $\lambda_{1,0} = \lambda_1 = 1$ over $\mathbb{F}_9$ ($n = 24 = 8 \cdot 3$, Table 5), four self-reciprocal orbits appear and $M = 3$ is odd, forcing $N_{\mathrm{SD}} = 0$ by (5.14).

Example 6.4 (Non-split affine algebra $\mathbb{F}_4 \times \mathbb{F}_{16}$ over $\mathbb{F}_4$)**.** Consider $\mathcal{A} = \mathbb{F}_4[X]/\langle X(X^2 + X + \omega)\rangle \cong K_1 \times K_2$ over $\mathbb{F}_4 = \mathbb{F}_2(\omega)$, where $K_1 = \mathbb{F}_4$ ($m_1 = 1$, $d_1 = 2$) and $K_2 = \mathbb{F}_{16}$ ($m_2 = 2$, $d_2 = 4$); since $m_2 = 2 > 1$, $\mathcal{A}$ lies strictly outside split-algebra frameworks such as [27]. Take $n = 10 = 5 \cdot 2^1$ ($n' = 5$, $M = 2$) and $k = 1$, so ambient $\mathbb{F}_4$-dimension is $W = 10(1 + 2) = 30$.

(a) At $\lambda_0 = (1, 1)$, $\lambda = \lambda_0^2 = (1, 1)$, $x^5 - 1$ has $\tau_{1,1}$-orbits $(1, 1), (2, 2)$ over $\mathbb{F}_4$ and splits completely over $\mathbb{F}_{16}$ into one fixed point $(x + 1)$ and one 4-cycle under $\alpha \mapsto \alpha^{-8} = \alpha^7 \equiv \alpha^2 \pmod 5$, contributing $\mathbb{F}_4$-weighted orbits $(1, 2)$ and $(4, 2)$. Theorem 4.6(b) enumerates all $3^8 = 6{,}561$ codes across 123 joint (κ, η_1) pairs, with $N_{\mathrm{LCD}} = 2^4 = 16$, $N_{\mathrm{SO}} = N_{\mathrm{DC}} = 2 \cdot 6 \cdot 2 \cdot 26 = 624$, $N_{\mathrm{SD}} = 1 \cdot 3 \cdot 1 \cdot 3 = 9$, $\mathbb{E}[\eta_1] = 23/3$, and $\mathrm{Var}(\eta_1) = 458/81$.

(b) At the mixed unit $\lambda_0 = (\omega, \beta)$, $\lambda = (\omega^2, \beta^2)$ ($\beta^4 + \beta + 1 = 0$ in $\mathbb{F}_{16}$), component K_1 remains compatible ($\omega^3 = 1$), whereas component K_2 has $\beta^{1+2^{4-1}} = \beta^9 \neq 1$ and $x^5 - \beta$ irreducible of degree 5 over $\mathbb{F}_{16}$ ($m_2 \deg f = 10$). By Theorem 4.6(a), $\mathscr{E}^{(2)}(U, Z) = \mathscr{E}^{(2)}_{K_1}(U, Z) \cdot (1 + U^{10} + U^{20})$ over all $27 \cdot 3 = 81$ codes, giving $4 \cdot 3 = 12$ LCD codes, 12 SO codes, 12 DC codes, and 0 self-dual codes.

Example 6.5 (Large-scale repeated-root families over $\mathbb{F}_{32}$)**.** Let $\mathcal{A} = \mathbb{F}_{32}$ ($p = 2$, $e = 5$, $m_1 = 1$), $k = 1$, $\lambda_{1,0} = \lambda_1 = 1$, and consider the 2-adic length tower $n_t = 31 \cdot 2^t$ ($n' = 31$, $M = 2^t$). Over $\mathbb{F}_{32}$, $x^{31} - 1$ splits into $N_F = 31$ linear factors partitioned into $(1, 1)$ and three orbits of type $(10, 1)$ ($|\mathcal{O}_1| = 4$).

- At $t = 1$ ($M = 2$, $n = 62$), Proposition 6.1 evaluates all $3^{31} = 617{,}673{,}396{,}283{,}947$ codes across 916 (κ, η_1) cells in under 0.05 s (Figure 9), yielding $N_{\mathrm{LCD}} = 16$, $N_{\mathrm{SO}} = N_{\mathrm{DC}} = 2 \cdot 3{,}281^3 = 70{,}639{,}674{,}082$, $N_{\mathrm{SD}} = 1 \cdot 3^3 = 27$, $\mathbb{E}[\eta_1] = 1/3 + 30(5/9) = 17$, $\mathrm{Var}(\eta_1) = 2/9 + 30(4/27) = 14/3$, and mid-rate peak conditional expectation

$$\mathbb{E}[\eta_1 \mid \kappa = 31] = \frac{937{,}013{,}336{,}624{,}311}{53{,}876{,}592{,}856{,}681} \approx 17.3918.$$

- At $t = 2$ ($M = 4$, $n = 124$), the code space grows to $5^{31} = 4{,}656{,}612{,}873{,}077{,}392{,}578{,}125$ codes across 3,531 cells (evaluated in under 0.3 s), with exact $N_{\mathrm{LCD}} = 16$ unchanged (probability $\approx 3.44 \times 10^{-21}$), $N_{\mathrm{SO}} = N_{\mathrm{DC}} = 3 \cdot 286{,}555^3 = 70{,}590{,}332{,}389{,}211{,}625$, $N_{\mathrm{SD}} = 1 \cdot 5^3 = 125$, $\mathbb{E}[\eta_1] = 184/5 = 36.8$, and $\mathrm{Var}(\eta_1) = 326/25 = 13.04$.

7. Conclusions

We have completely resolved the joint enumeration of code dimension $\kappa(C)$ and k-Galois hull dimension $\eta_k(C)$ across all $(M + 1)^{N_F}$ labeled $\mathcal{A}$-linear λ-constacyclic codes of arbitrary length $n = n'p^t$ ($M = p^t$, $\gcd(n', p) = 1$) over any finite commutative square-free affine algebra $\mathcal{A} \cong \prod_{s=1}^{N} \mathbb{F}_{q^{m_s}}$ ($m_s \geq 1$) and any unit $\lambda \in \mathcal{A}^*$. Passing to the unique p^t-th roots $\lambda_{s,0} \in K_s^*$

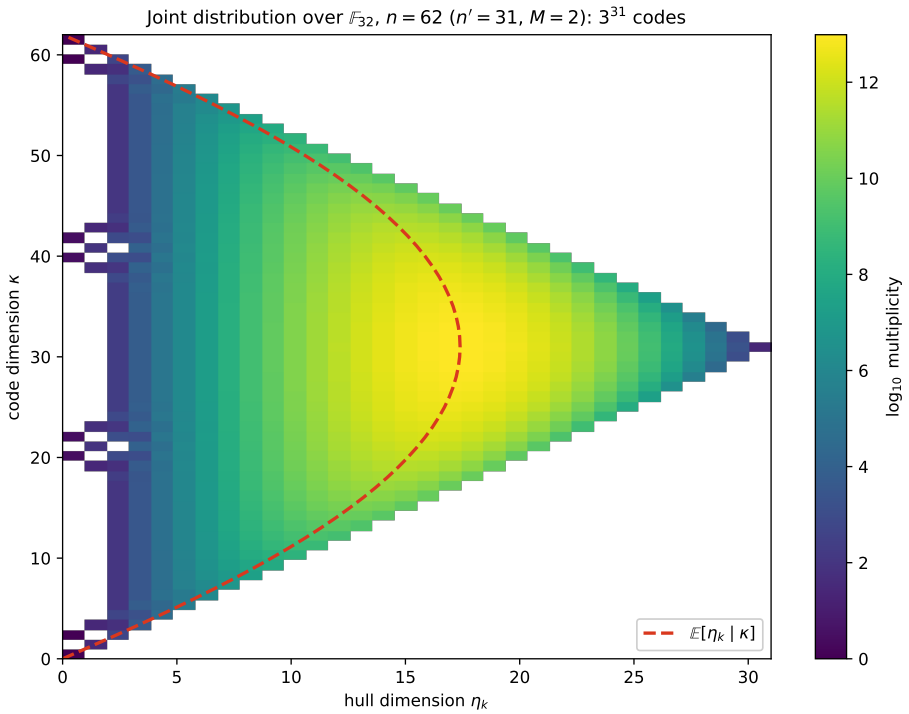


Figure 9: Exact joint log-multiplicity distribution across all $3^{31} = 617{,}673{,}396{,}283{,}947$ cyclic codes of repeated-root length $n = 62$ ($n' = 31$, $M = 2$) over $\mathbb{F}_{32}$, with exact conditional expectation curve $\mathbb{E}[\eta_1 \mid \kappa]$ overlaid.

preserves twist compatibility unchanged ($\lambda_{s,0}^{1+p^{d_s-k}} = 1$), collapses all incompatible components to trivial hulls, and turns $\dim_{K_s} \mathrm{Hull}_k(C_s(e))$ on compatible components into a sum of cyclic edge minima $\min\{e_{O,i}, M - e_{O,i+1}\}$ governed by universal $(M+1) \times (M+1)$ transfer matrices $T_{w_O}^{(M)}(U, Z)$. This exact algebraic machinery answers the open counting problem posed by Debnath et al. [27, p. 23] without split-algebra or order-coprimality restrictions, contains the simple-root theory ($M = 1$) as an immediate special case, proves p-power invariance $2^{\sum_s |\mathcal{O}_s|}$ of compatible k-Galois LCD code counts, establishes unconditionally the exact single-orbit support trapezoid and global Minkowski hull spectrum, yields closed rational mean, variance, and conditional-moment laws alongside exact self-orthogonal, dual-containing, and self-dual code counts, and runs in polynomial time even on code spaces exceeding 10^{21}.

Three natural open directions emerge directly from these results:

1. *Simultaneous ring ramification and repeated-root lengths.* When $\mathcal{A}$ itself has nilpotent radical (for instance chain rings $\mathbb{F}_{p^m}[u]/\langle u^r \rangle$ or general finite commutative local rings [17, 18, 21, 23, 32, 38]) and $p \mid n$ simultaneously, ambient rings cease to be principal ideal rings and exponents are replaced by torsion filtrations; extending transfer-matrix enumeration to that regime remains open.
2. *Multi-twisted and quasi-twisted generalisations.* Extending our $(M+1)$-state transfer operators from single-modulus constacyclic codes to double cyclic, quasi-constacyclic, and multi-twisted code families [22, 24–26], where $\mathcal{A}[x]/\langle x^n - \lambda \rangle$-ideals become rank-$r$ submodules of $\bigoplus_{j=1}^{r} \mathcal{A}[x]/\langle x^{n_j} - \lambda_j \rangle$, involves non-abelian Grassmannian / Hermite-normal-form local factors rather than scalar chains.

3. *Minimum-distance stratification inside fixed* (κ, η_k) *cells.* Combining exact (κ, η_k) cell multiplicities with torsion minimum-distance bounds [30, 38] to construct optimal repeated-root EAQECCs with prescribed ebit count $c = \kappa - \eta_k$ [7–9] is an appealing direction for quantum code design.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

All exact coefficient tables, figures, and self-contained standard-library Python verification scripts reproducing every calculation in this paper are provided in the accompanying supplementary material package.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Acknowledgements

The authors thank the anonymous referees for their careful reading and constructive suggestions.

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